LABSAFE – LABORATORY CONSUMABLE INVENTORY & EXPIRY MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

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240701164

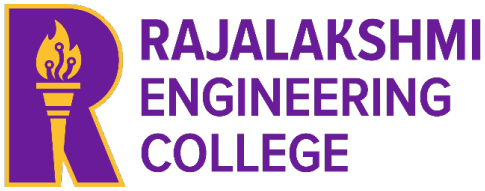
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CB23F35 –
LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING

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RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled **"LabSafe – Laboratory Consumable Inventory & Expiry Management System"** is the bonafide work of HARIKRISHNAN EP (240701164), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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INTERNAL EXAMINER	**EXTERNAL EXAMINER**

DECLARATION

I, HARIKRISHNAN EP (240701164), hereby declare that the project report titled "**LabSafe – Laboratory Consumable Inventory & Expiry Management System**", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai Date:	**SIGNATURE OF THE STUDENT** HARIKRISHNAN EP 240701164

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HARIKRISHNAN EP

ABSTRACT

Educational, research and testing laboratories depend on consumables such as reagents, chemicals, glassware and disposable items. These items arrive in batches, are stored in different locations, carry expiry dates and are issued to many users. When purchases, issues and expiry dates are recorded manually in registers or spreadsheets, stock quantities become difficult to keep up to date, low-stock and expired items can go unnoticed, and reports must be compiled by hand.

LabSafe is a laboratory consumable inventory and expiry management application developed on the Zoho Creator low-code platform. It consists of six main forms — Category_Master, Supplier_Master, Consumable_Master, Stock_Batch, Stock_Purchase and Stock_Issue — and two child subforms, Purchase_Items_Form and Issue_Items_Form, connected through lookup fields and parent–child relationships. Five form workflows written with Deluge scripting validate stock issues, deduct issued quantities from batch stock, increase batch stock for matching purchase records, assign a batch status (Available, Expiring Soon, Low Stock, Expired or Depleted) and send an expiry email notification. Five custom reports, six default list reports and a dashboard named LabSafe_Dashboard give an overview of batch stock, low-stock items, expired items and issue transactions. Admin and Lab Staff roles were defined for the application.

The application was tested for record creation, insufficient-stock validation, quantity deduction after issue, quantity update after a purchase against an existing batch, expiry status update, email delivery, and the display of reports, the stock status chart and the dashboard; all fourteen recorded tests passed. The report also states the limitations that remain: new batches are not created automatically from purchases, the expiry email may fire on any batch add or edit, automatic daily expiry checking, complete role-based access enforcement, production deployment and performance have not been verified, and no external integration, AI forecasting or separate mobile application has been developed. These form the basis of the proposed future enhancements.

PROJECT AT A GLANCE

Attribute	Details
Project Title	LabSafe – Laboratory Consumable Inventory & Expiry Management System
Application Domain	Laboratory inventory management
Student Name & Register No.	HARIKRISHNAN EP – 240701164
Zoho Creator Edition	Free
Application Name (Link Name)	LabSafe (labsafe)
Published / Live URL	https://creatorapp.zoho.in/harikrishnan.ep.2024.cse1207/labsafe#Form:Category_Master
Demo Credentials (per role)	Admin, | Role 2: Lab Staff
Number of Forms / Subforms	6 main forms / 2 child subforms (Purchase_Items_Form, Issue_Items_Form)
Number of Reports (by type)	Custom reports: 5 (Stock_Batch_Report, Low_Stock_Report, Expired_Items_Report, Stock_Issue_Report, Stock_Status_Chart – chart report). Default list reports: 6. Kanban / Calendar / Timeline / Map: none
Number of Pages / Dashboards	1 dashboard (LabSafe_Dashboard)
Number of Workflows	Form workflows: 5 (Update_Stock_After_Issue, Validate_Stock_Before_Issue, Expiry_Alert, Expiry_Email_Notification, Update_Stock_After_Purchase). Schedule: 0. Approval: 0. Blueprint: 0. Batch: 0
Deluge Functions / Custom Actions	Deluge logic in three areas (issue validation, quantity update, expiry/status update) plus email notification;
Integrations	Zoho Creator email notification
Roles & Profiles Defined	Admin, Lab Staff (permission configuration inspected and configured; complete enforcement not fully verified)
Environments Used	Development
Date of Production Deployment	31/10/2026

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CO	Course Outcome
CRUD	Create, Read, Update, Delete
DFD	Data Flow Diagram
ER	Entity–Relationship
FR	Functional Requirement
LCAP	Low-Code Application Platform
NFR	Non-Functional Requirement
RBAC	Role-Based Access Control
SDLC	Software Development Life Cycle
TC	Test Case
UAT	User Acceptance Testing
UI	User Interface
UC	Use Case

CHAPTER 1

INTRODUCTION

1.1 Background

Laboratories in educational institutions, research centres and testing facilities use a wide range of consumables, including chemicals, reagents, glassware, disposable plasticware and similar items. These consumables are bought from suppliers in batches, stored in designated locations under specified storage conditions, and issued in small quantities to students, researchers and technicians for experiments and practical sessions.

Two characteristics make consumable management more demanding than ordinary stock keeping. First, many consumables have a manufacturing date and an expiry date, so stock must be tracked at the level of individual batches rather than only at the level of the item. Second, every consumable has a minimum level below which it must be re-ordered so that laboratory work is not interrupted. Staff therefore need to know, at any time, how much of each batch remains, which batches are close to expiry or already expired, and which items are running low.

Where this information is kept in paper registers or spreadsheets, each purchase and each issue must be written down and the balance recalculated by hand. LabSafe was developed to bring these activities into a single Zoho Creator application so that consumable details, supplier information, stock batches, purchase transactions and stock issue transactions are maintained in one place and the stock balance and batch status are updated by workflows.

1.2 Problem Statement

Manual or spreadsheet-based recording of laboratory consumables makes it difficult to maintain accurate batch-level stock, to detect expired and low-stock items in time, and to prevent stock from being issued in quantities larger than what is available. The following specific problems were identified:

- Stock balances depend on manual calculation after every purchase and issue, which is slow and error-prone.
- Expired, expiring and low-stock items are noticed only when somebody inspects the records or the shelf.
- Nothing prevents an issue entry from exceeding the quantity actually available in a batch.
- Supplier details, consumable details and transactions are scattered across separate files or registers.
- Reports on current stock, expired items and issue history must be prepared manually whenever they are needed.

1.3 Objectives

- To design a data model and forms for categories, suppliers, consumables, stock batches, purchases and issues, linked through lookup fields and subforms.
- To automate stock deduction after issue, stock increase after purchase of an existing batch, and batch-status calculation using Deluge scripting and form workflows.
- To validate stock issue entries so that an unselected batch, a zero or negative quantity, and a quantity greater than the available stock are rejected.
- To provide reports and a dashboard that show batch stock, low-stock items, expired items and issue transactions.
- To send an email notification containing batch number, expiry date, available quantity and batch status when a stock batch is added or edited.
- To define Admin and Lab Staff roles for the application and to document honestly which features were verified and which remain limited.

1.4 Scope of the Project

In scope. The project covers the following modules, all implemented on Zoho Creator and accessed through a web browser:

- Master data maintenance: Category_Master, Supplier_Master and Consumable_Master.
- Batch-level stock records in Stock_Batch, with manufacturing date, expiry date, initial and available quantity and batch status.
- Stock_Purchase transactions with a Purchase_Items subform, and Stock_Issue transactions with an Issue_Items subform.
- Form workflows for issue validation, stock deduction, purchase quantity update, expiry/stock status update and expiry email notification.
- Five custom reports, six default list reports and the LabSafe_Dashboard.
- Two roles: Admin and Lab Staff.

Out of scope:

The following were not implemented and are discussed as limitations and future enhancements:

- External API integrations.
- AI-based stock forecasting or expiry prediction.
- A separately developed mobile application.
- Automatic purchase approval and supplier payment.
- Automatic creation of a new batch from a purchase.
- A verified daily scheduled expiry check.
- Production and multi-environment deployment.
- Complete role-specific access enforcement and performance benchmarking were not verified.

1.5 Justification for a Low-Code Approach

Low-code application platforms allow business applications to be built mainly through visual configuration, with a small amount of scripting where business rules require it . Industry analysts treat enterprise low-code platforms as a distinct and

growing product category. For LabSafe the approach is suitable for the following reasons.

- **Speed and fit.** Forms, lookup fields, subforms, reports and a dashboard are configured visually in Zoho Creator, so the data model could be built and changed quickly as requirements became clear.
- **Focused scripting.** Only the business rules that need logic — stock validation, quantity update and status calculation — were written in Deluge; the rest is configuration.
- **Built-in hosting and access.** The platform hosts the application and provides user, role and permission settings, so no separate server, database or front-end code had to be developed.
- **Maintainability.** Laboratory staff with basic training can adjust fields, reports and workflow conditions without rewriting a code base.

The trade-offs of the approach — dependence on the platform and the limits of what can be configured — are discussed in Section 8.4.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements. The appendices contain the feature coverage checklist, course outcome mapping, Deluge logic, user manual, evaluation rubric and viva-voce record sheet.

CHAPTER 2

LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In the existing arrangement, laboratory consumables are managed through manual records such as stock registers, issue registers or spreadsheets. The description below is a generalised account of that manual process, which LabSafe is intended to replace; it is not a measured study of any one laboratory.

- When consumables are received from a supplier, the details are written into a register or spreadsheet.
- The items are placed in the laboratory store; the storage location and conditions are known to the person in charge.
- When items are issued, the quantity is noted manually and the balance is recalculated.
- Stock is checked periodically by counting items or reading records; expiry dates and minimum levels are checked by inspection.
- Reports on stock or issues are prepared by collecting the entries by hand.

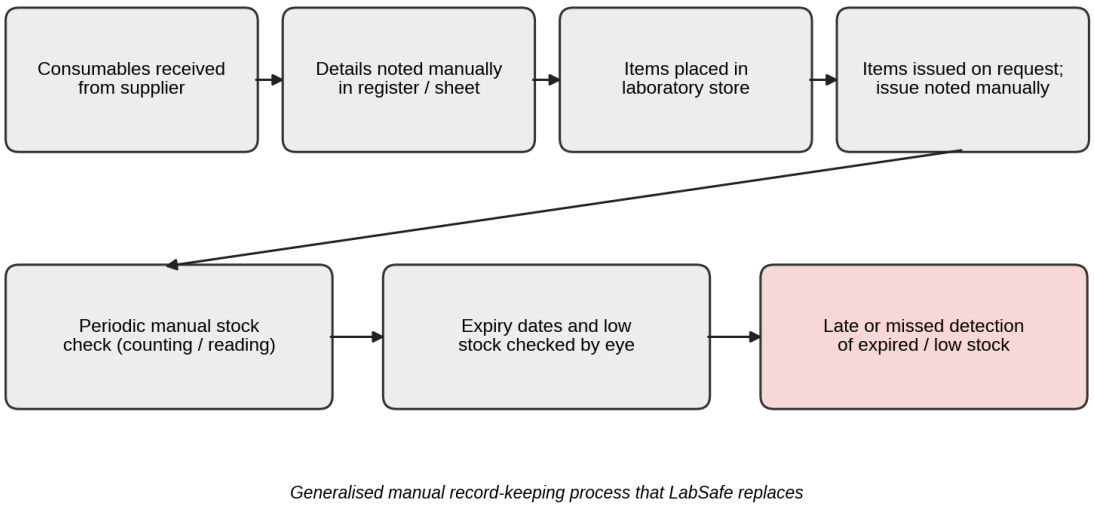


Figure 2.1: As-Is Process Flow of the Existing Manual System

2.2 Limitations of the Existing System

- Stock balances are recalculated manually, so errors accumulate and the recorded balance may differ from the physical stock.
- Expired and low-stock items are detected late because nothing alerts the staff.
- Issues larger than the available quantity are not prevented at the time of entry.
- Batch-level information (batch number, manufacturing date, expiry date) is difficult to keep consistent with item-level records.
- Supplier, consumable and transaction details are scattered, which makes searching and reporting slow.
- Information is not available to several authorised users at the same time and cannot be filtered on demand.

2.3 Low-Code versus Traditional Development

Table 2.1 compares the two development approaches for the LabSafe application. The comparison is qualitative; no development-effort figures were measured for a traditionally coded version, so none are claimed.

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Not estimated; separate design and coding of database, back end and user interface would be needed	Estimated 1–2 weeks
Skills required	Programming, database and web development	Data modelling, form and report configuration, basic Deluge scripting
Hosting & infrastructure	Server, database and deployment to be arranged separately	Hosted by the Zoho Creator platform
Mobile application	Separate mobile development would be required	Not developed for LabSafe (out of scope)
Security & access control	Authentication and authorisation to be coded	Platform user, role and permission settings; complete enforcement for LabSafe not fully verified
Maintenance & change	Code changes, testing and redeployment	Fields, reports and workflows modified in the builder
Customisation limits	Few limits	Bound by what the platform and Deluge allow
Cost	Development and hosting costs	Platform subscription or education access

2.4 Literature Survey

The sources reviewed are the course reference text on low-code development, an analyst report on the low-code platform market, and the official Zoho Creator documentation used to configure the application. Their relevance to this project is summarised in Table 2.2. The details of publication year and access date are to be completed from the original sources.

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	W. Rosam and O. Kalnichevski, BPB Publications [1]	Low-Code Development: A Practical Guide	Course reference text on low-code development principles and practice	Conceptual basis for the low-code approach in Section 1.5 and Chapter 8
2	Gartner, Inc., 2024 [2]	Magic Quadrant for Enterprise Low-Code Application Platforms	Analyst evaluation of enterprise low-code application platforms	Industry context for low-code platform adoption
3	Zoho Corporation [3]	Zoho Creator Help Documentation	Official description of forms, lookup fields, subforms, reports, workflows and pages	Reference for configuring LabSafe forms, reports and dashboard
4	Zoho Corporation [4]	Deluge Script Reference Guide	Official reference for the Deluge scripting language	Reference for validation, fetch, update and notification logic
5	Zoho Corporation [5]	Zoho Creator Learning Centre	Learning resources for building Zoho Creator applications	Support for learning platform features during development

2.5 Emerging Trends in Low-Code Development

Several trends are commonly associated with low-code platforms: citizen development, in which domain users build or adapt applications themselves; hyper-automation, in which workflows automate routine business steps; AI-assisted application building and analytics; and composable applications assembled from reusable components. These trends are discussed in the course material and in analyst literature on the sector [1], [2].

LabSafe reflects the first two. A domain-focused application was assembled mainly through configuration by a student developer, and routine stock steps — validation, quantity update, status calculation and email alerts — were automated through workflows. The application does **not** use AI-assisted features, predictive analytics or external integrations; AI-based forecasting is therefore listed as a future enhancement rather than as an implemented feature.

2.6 Proposed System

The proposed system is the LabSafe application on Zoho Creator. Table 2.3 compares it with the existing manual system and shows how each limitation listed in Section 2.2 is addressed.

Table 2.3: Comparison of the Existing and Proposed Systems

Aspect	Existing Manual System	Proposed LabSafe Application
Data storage	Registers or spreadsheets, scattered	Six main forms with two child subforms, linked by lookup fields
Stock quantity	Calculated by hand after each transaction	Available_Quantity reduced after a valid issue and increased after a purchase of a matching existing batch by workflows
Issue control	No check at entry	Validation blocks an unselected batch, zero or negative quantity, and quantity above available stock
Expiry and low stock	Found by inspection	Batch_Status calculated as Expired, Depleted, Low Stock, Expiring Soon or Available when a batch is added or edited
Alerts	None	Email notification with batch number, expiry date, available quantity and status on batch add/edit
Reports	Prepared manually	Stock_Batch_Report, Low_Stock_Report, Expired_Items_Report, Stock_Issue_Report, Stock_Status_Chart and default list reports
Overview	None	LabSafe_Dashboard
Known gaps	—	No automatic new-batch creation from purchase; email may fire on any add/edit; daily check without edit and full role enforcement not verified

Chapters 3 and 4 translate this proposal into requirements and a design.

CHAPTER 3

REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Admin	Person responsible for managing laboratory inventory and the application	Manage inventory records, transactions, reports and application settings
Lab Staff	Laboratory technician or staff member who uses consumables	Access permitted inventory functions and record stock issues
Email recipient	Person who receives the expiry email notification	Review batch number, expiry date, quantity and status in the email

The Admin and Lab Staff roles were created in the application. Role-specific permissions were inspected and configured, but complete role-specific access enforcement was not fully verified (see Sections 5.8 and 7.2).

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall maintain laboratory consumable categories (name, code, description, status).	Admin	High
FR-02	The system shall maintain supplier details (name, code, contact person, phone, email, address, status).	Admin	High
FR-03	The system shall maintain consumables with category and supplier lookups, unit, minimum stock level, storage location and conditions, and status.	Admin	High
FR-04	The system shall maintain stock batches with batch number, consumable, dates, initial and available quantity, storage location and batch status.	Admin	High
FR-05	The system shall record stock purchases with multiple purchased items through a Purchase_Items subform.	Admin	High
FR-06	After a successful purchase, the system shall increase Available_Quantity of an existing Stock_Batch matching the batch number and consumable.	Admin	High
FR-07	The system shall record stock issues with multiple issued items through an Issue_Items subform.	Admin, Lab Staff	High
FR-08	The system shall block a stock issue if no batch is selected, the quantity is zero or negative, or the quantity exceeds available stock.	Admin, Lab Staff	High
FR-09	After a successful issue, the system shall deduct the issued quantity from the selected batch's Available_Quantity.	Admin, Lab Staff	High
FR-10	The system shall set Batch_Status (Depleted, Expired, Low Stock, Expiring Soon, Available) when a batch is added or edited.	System	High
FR-11	The system shall send an email notification with batch number, expiry date, available quantity and status when a batch is added or edited.	System	Medium
FR-12	The system shall provide reports for stock batches, low-stock items, expired items and stock issues.	Admin, Lab Staff	High
FR-13	The system shall provide a stock status chart and the LabSafe_Dashboard.	Admin, Lab Staff	Medium
FR-14	The system shall define Admin and Lab Staff roles with permissions (enforcement not fully verified).	Admin	Medium

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement	Verification Status
NFR-01	Performance	Forms and reports should respond acceptably for laboratory-scale data volumes	Not benchmarked
NFR-02	Security	Users should access only the functions permitted to their role	Roles configured; full enforcement not verified
NFR-03	Usability	Users should record purchases and issues through simple forms and subforms in a web browser	Observed during testing; no formal usability study
NFR-04	Data integrity	Stock must not be issued beyond available quantity	Verified (TC-07)
NFR-05	Maintainability	Fields, reports and workflows should be changeable in the Zoho Creator builder without rewriting code	Supported by platform design
NFR-06	Availability	Application availability depends on the hosting platform	Not measured

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Application components	Zoho Creator forms, reports, pages/dashboard, workflows, email notifications
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
Client hardware	Computer or laptop with a modern processor, 4 GB RAM or above, and an internet connection
External services	None; no external APIs, third-party databases or AI models are used
Mobile	No separate mobile application developed

3.5 Use Case Model

Figure 3.1 shows the use cases of LabSafe. The Admin can use all functions. The Lab Staff role is intended to record stock issues and view reports and the dashboard; this is

the intended design, and role enforcement was not fully verified. Three key use cases are described in Tables 3.5 to 3.7.

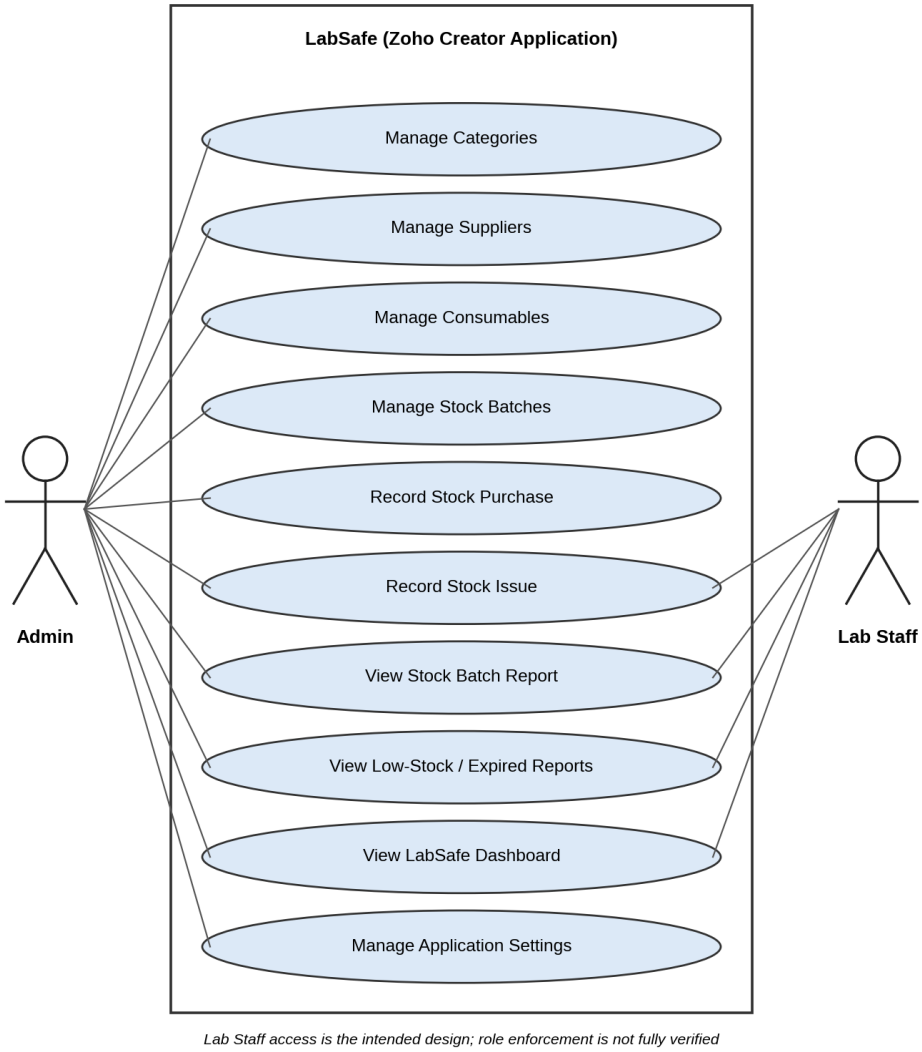


Figure 3.1: Use Case Diagram of LabSafe

Table 3.5: Use Case Description – Record Stock Issue

Item	Description
Use Case ID	UC-01
Use Case Name	Record Stock Issue
Actor(s)	Lab Staff, Admin
Pre-condition	Consumable and Stock_Batch records exist with available quantity
Main Flow	1. Open the Stock_Issue form. 2. Enter issue details. 3. Add Issue_Items rows with consumable, batch, quantity and remarks. 4. Submit. 5. Validation passes. 6. Record is saved. 7. Available_Quantity of each selected batch is reduced.
Alternate Flow	If no batch is selected, quantity is zero or negative, or quantity exceeds available stock, submission is blocked and the user corrects the entry.
Post-condition	Issue recorded; batch stock reduced
Related FR	FR-07, FR-08, FR-09

Table 3.6: Use Case Description – Record Stock Purchase

Item	Description
Use Case ID	UC-02
Use Case Name	Record Stock Purchase
Actor(s)	Admin
Pre-condition	Supplier and Consumable records exist
Main Flow	1. Open Stock_Purchase. 2. Enter purchase date, supplier, invoice number, received by and notes. 3. Add Purchase_Items rows (consumable, batch number, quantity, dates, unit cost). 4. Submit. 5. For each item, a Stock_Batch with the same batch number and consumable is searched.
Alternate Flow	If no matching batch exists, the configured workflow does not create one; the batch must be created in Stock_Batch.
Post-condition	Purchase recorded; Available_Quantity increased for a matching existing batch
Related FR	FR-05, FR-06

Table 3.7: Use Case Description – Update Batch Status and Notify

Item	Description
Use Case ID	UC-03
Use Case Name	Update Batch Status and Send Notification
Actor(s)	Admin (creates or edits batch); system
Pre-condition	Consumable has a minimum stock level; batch has expiry date and quantity
Main Flow	1. A Stock_Batch is added or edited. 2. Expiry_Alert evaluates quantity, expiry date and minimum stock level. 3. Batch_Status is updated. 4. Expiry_Email_Notification sends the email.
Alternate Flow	The email may be sent for any successful add or edit, not only when an expiry condition first occurs.
Post-condition	Batch_Status reflects the configured conditions; email sent
Related FR	FR-10, FR-11

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

LabSafe follows the layered structure of a Zoho Creator application (Figure 4.1). Users with the Admin or Lab Staff role access the application through a web browser. The application layer contains six main forms with two child subforms, five form workflows with Deluge scripts, five custom reports with six default list reports, and the LabSafe_Dashboard. Roles and permissions govern access, and the email notification feature sends the expiry alert. Form data is stored in the data layer managed by Zoho Creator.

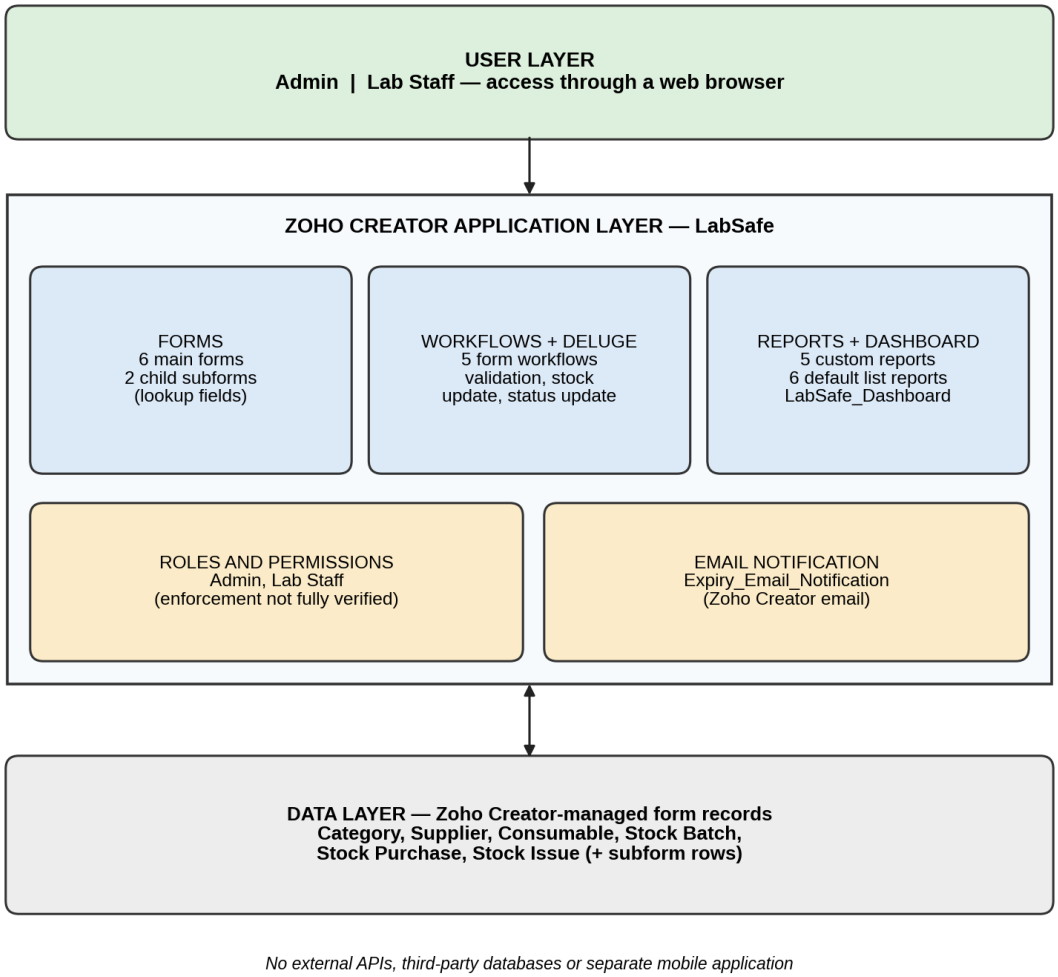


Figure 4.1: System Architecture of LabSafe

The architecture has no integration layer: no external APIs, third-party databases or AI models are used, and no separate mobile application is built.

4.2 Data Model

The data model consists of six main forms and two child subforms. Relationships are realised through **lookup fields** (a field in one form that selects a record of another form) and through **subforms** (child rows embedded in a parent form). Zoho Creator assigns each record a system-generated record ID, so no manually designed relational schema or database IDs are defined; the “PK” label in Figure 4.2 denotes this system record ID and “FK (Lookup)” denotes a lookup field.

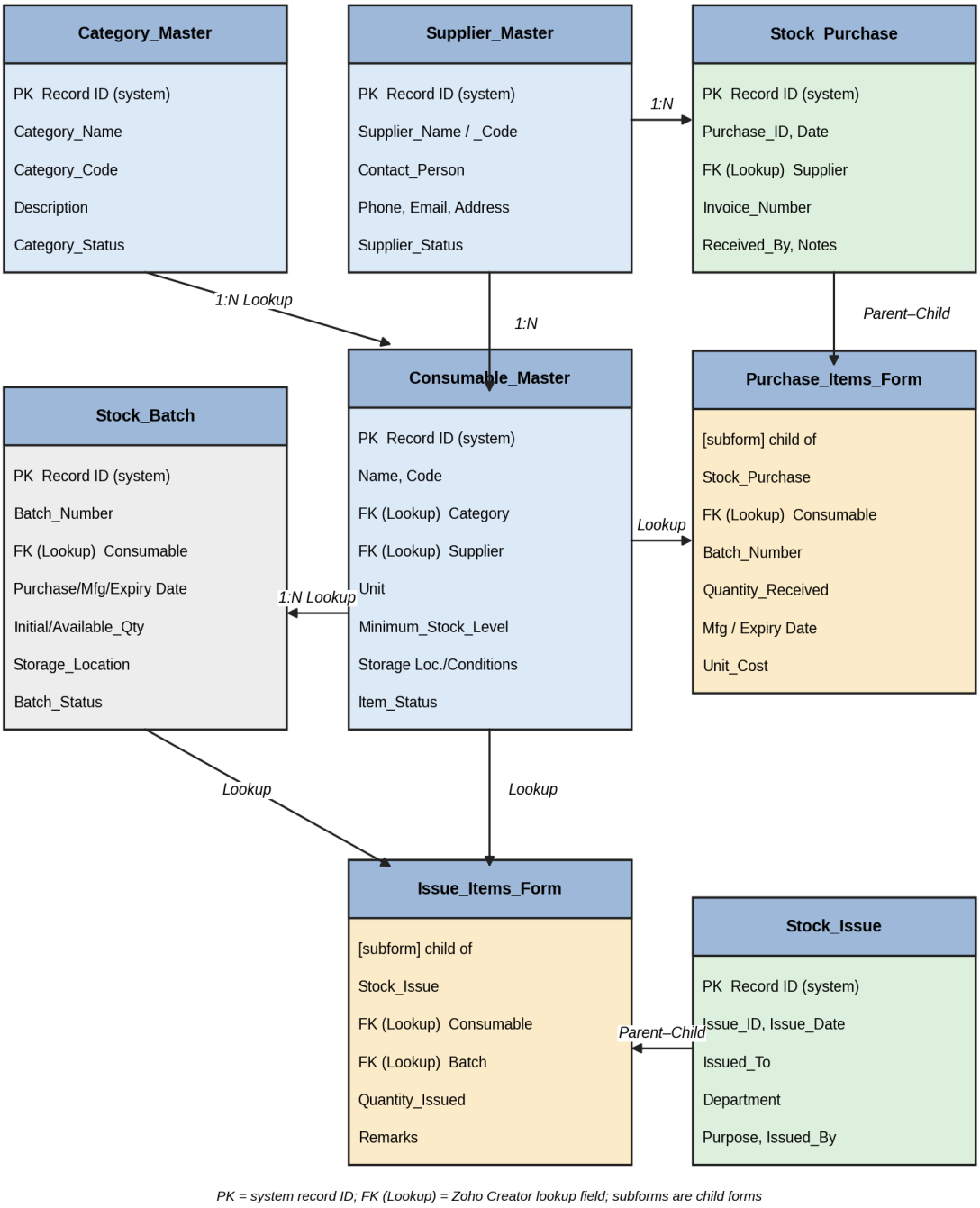


Figure 4.2: Entity–Relationship Diagram of LabSafe

The two child subforms, Purchase_Items_Form and Issue_Items_Form, are **not independent business modules**: they exist only to hold the multiple item rows of a Stock_Purchase or Stock_Issue transaction.

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Category_Master	Category_Name, Category_Code, Category_Status	Consumable_Master	One-to-Many	Lookup (Category) in Consumable_Master
Supplier_Master	Supplier_Name, Supplier_Code, Supplier_Status	Consumable_Master	One-to-Many	Lookup (Supplier) in Consumable_Master
Supplier_Master	Supplier_Name, Supplier_Code	Stock_Purchase	One-to-Many	Lookup (Supplier) in Stock_Purchase
Consumable_Master	Consumable_Name, Consumable_Code, Minimum_Stock_Level	Stock_Batch	One-to-Many	Lookup (Consumable) in Stock_Batch
Stock_Purchase	Purchase_ID, Purchase_Date, Invoice_Number	Purchase_Items_Form	Parent–Child	Subform (Purchase_Items)
Consumable_Master	—	Purchase_Items_Form	Lookup	Lookup (Consumable) in subform
Stock_Issue	Issue_ID, Issue_Date, Issued_To	Issue_Items_Form	Parent–Child	Subform (Issue_Items)
Consumable_Master	—	Issue_Items_Form	Lookup	Lookup (Consumable) in subform
Stock_Batch	Batch_Number, Expiry_Date, Available_Quantity	Issue_Items_Form	Lookup	Lookup (Batch) in subform

4.3 Form Design

Tables 4.2 to 4.9 present the fields associated with the eight forms used in the LabSafe application, based on the project implementation records. The field types are identified according to their intended data-entry and relationship functions in Zoho Creator. Field link names are provided where confirmed; any unverified link names are marked for confirmation against the actual form configuration. Mandatory settings and field-level validation rules are documented only where they have been verified in the application.

Table 4.2: Field Specification – Category_Master Form

Field Label	Link Name	Field Type	Remarks
Category Name	Category_Name	Single Line	Name of the category
Category Code	Category_Code	Single Line	Short code
Description	Description	Multi Line	Optional description
Category Status	Category_Status	Drop Down	Active / inactive state

Table 4.3: Field Specification – Supplier_Master Form

Field Label	Link Name	Field Type	Remarks
Supplier Name	Supplier_Name	Single Line	Stores the name of the supplier
Supplier Code	Supplier_Code	Single Line	Unique identification code for each supplier
Contact Person	Contact_Person	Single Line	Name of the supplier's contact person
Phone Number	Phone_Number	Phone	Stores the supplier's contact number
Email	Email	Email	Stores the supplier's email address
Address	Address	Multi Line	Stores the supplier's complete address
Supplier Status	Supplier_Status	Drop Down	Indicates whether the supplier is Active, Inactive or Other

Table 4.4: Field Specification – Consumable_Master Form

Field Label	Link Name	Field Type	Remarks
Consumable Name	Consumable_Name	Single Line	Name of the consumable
Consumable Code	Consumable_Code	Single Line	Unique consumable code
Category	Category	Lookup	Linked category
Unit	Unit	Single Line / Drop Down	Unit of measurement
Minimum Stock Level	Minimum_Stock_Level	Number	Minimum required quantity
Storage Location	Storage_Location	Single Line	Storage place
Storage Conditions	Storage_Conditions	Single / Multi Line	Storage requirements
Supplier	Supplier	Lookup	Linked supplier
Item Status	Item_Status	Drop Down	Current item status

Table 4.5: Field Specification – Stock_Batch Form

Field Label	Link Name	Field Type	Remarks
Batch Number	Batch_Number	Single Line	Unique batch number
Consumable	Consumable	Lookup	Linked consumable
Purchase Date	Purchase_Date	Date	Date of purchase
Manufacturing Date	Manufacturing_Date	Date	Manufacturing date
Expiry Date	Expiry_Date	Date	Batch expiry date
Initial Quantity	Initial_Quantity	Number	Original quantity
Available Quantity	Available_Quantity	Number	Remaining quantity
Storage Location	Storage_Location	Single Line	Batch storage place
Batch Status	Batch_Status	Drop Down / Single Line	Current batch status

Table 4.6: Field Specification – Stock_Purchase Form

Field Label	Link Name	Field Type	Remarks
Purchase ID	Purchase_ID	Single Line / Auto Number	Purchase identification
Purchase Date	Purchase_Date	Date	Date of purchase
Supplier	Supplier	Lookup	Linked supplier
Invoice Number	Invoice_Number	Single Line	Invoice reference
Received By	Received_By	Single Line	Person receiving stock
Purchase Notes	Purchase_Notes	Multi Line	Additional purchase details
Purchase_Items	Purchase_Items	Subform	Purchased item details

Table 4.7: Field Specification – Stock_Issue Form

Field Label	Link Name	Field Type	Remarks
Issue ID	Issue_ID	Single Line / Auto Number	Issue identification
Issue Date	Issue_Date	Date	Date of issue
Issued To	Issued_To	Single Line	Recipient name
Department	Department	Single Line / Drop Down	Receiving department
Purpose	Purpose	Multi Line	Reason for issue
Issued By	Issued_By	Single Line	Person issuing stock
Issue_Items	Issue_Items	Subform	Issued item details

Table 4.8: Field Specification – Purchase_Items_Form (Child Subform)

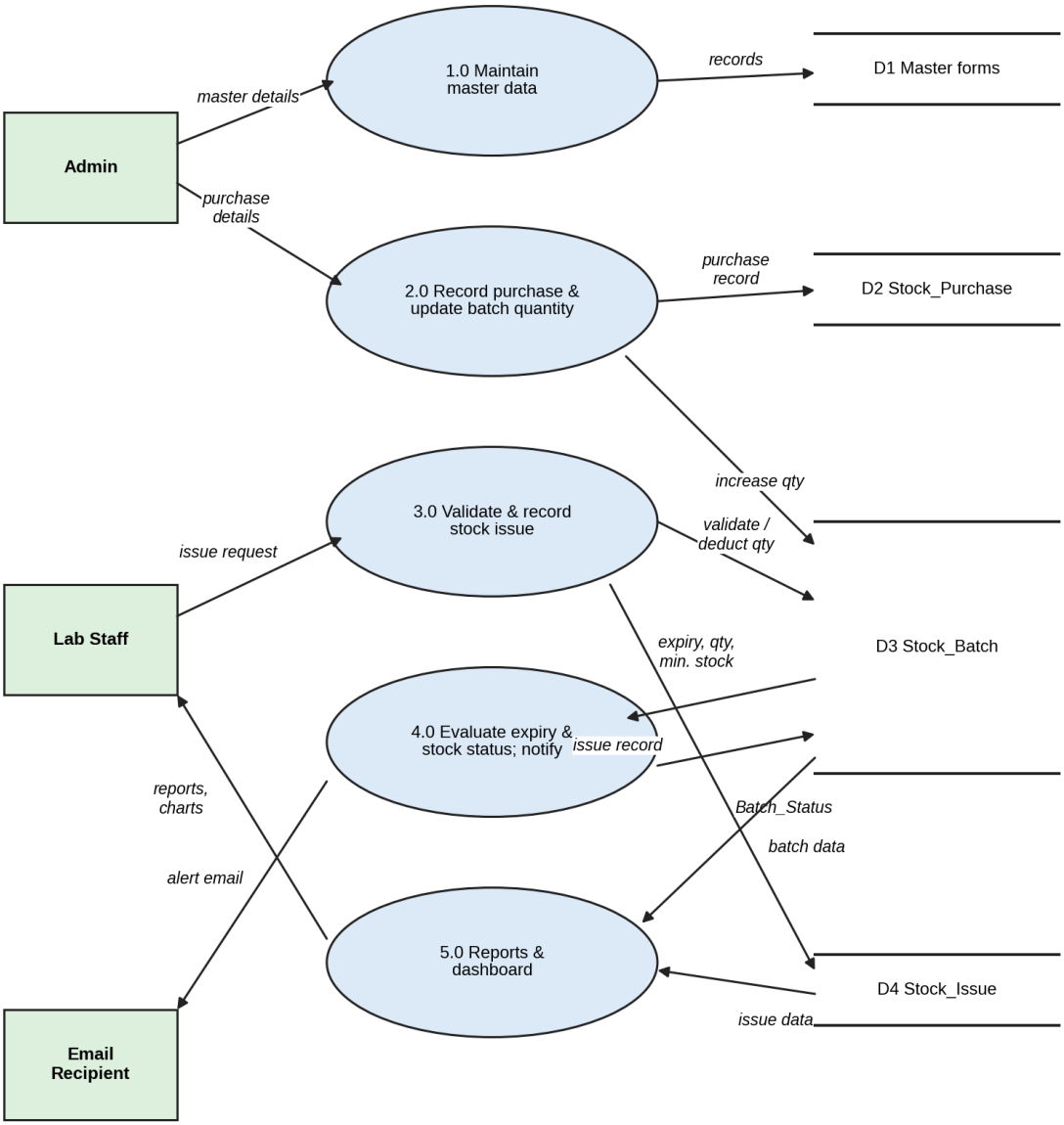
Field Label	Link Name	Field Type	Remarks
Consumable	Consumable	Lookup	Linked consumable
Batch Number	Batch_Number	Single Line	Batch identification
Quantity Received	Quantity_Received	Number	Quantity purchased
Manufacturing Date	Manufacturing_Date	Date	Manufacturing date
Expiry Date	Expiry_Date	Date	Expiry date
Unit Cost	Unit_Cost	Decimal / Currency	Cost per unit

Table 4.9: Field Specification – Issue_Items_Form (Child Subform)

Field Label	Link Name	Field Type	Remarks
Consumable	Consumable	Lookup	Linked consumable
Batch	Batch	Lookup	Selected stock batch
Quantity Issued	Quantity_Issued	Number	Quantity issued
Remarks	Remarks	Single / Multi Line	Additional comments

4.4 Process and Workflow Design

Three processes are automated: stock issue (validation followed by deduction), stock purchase (quantity update for an existing matching batch) and batch expiry/status evaluation with an email notification. Figures 4.3 to 4.6 show the data flow and the three workflows; each box marked “Workflow” runs a Deluge script.



Reports also read D1 and D2 (arrows omitted for clarity)

Figure 4.3: Data Flow Diagram of LabSafe

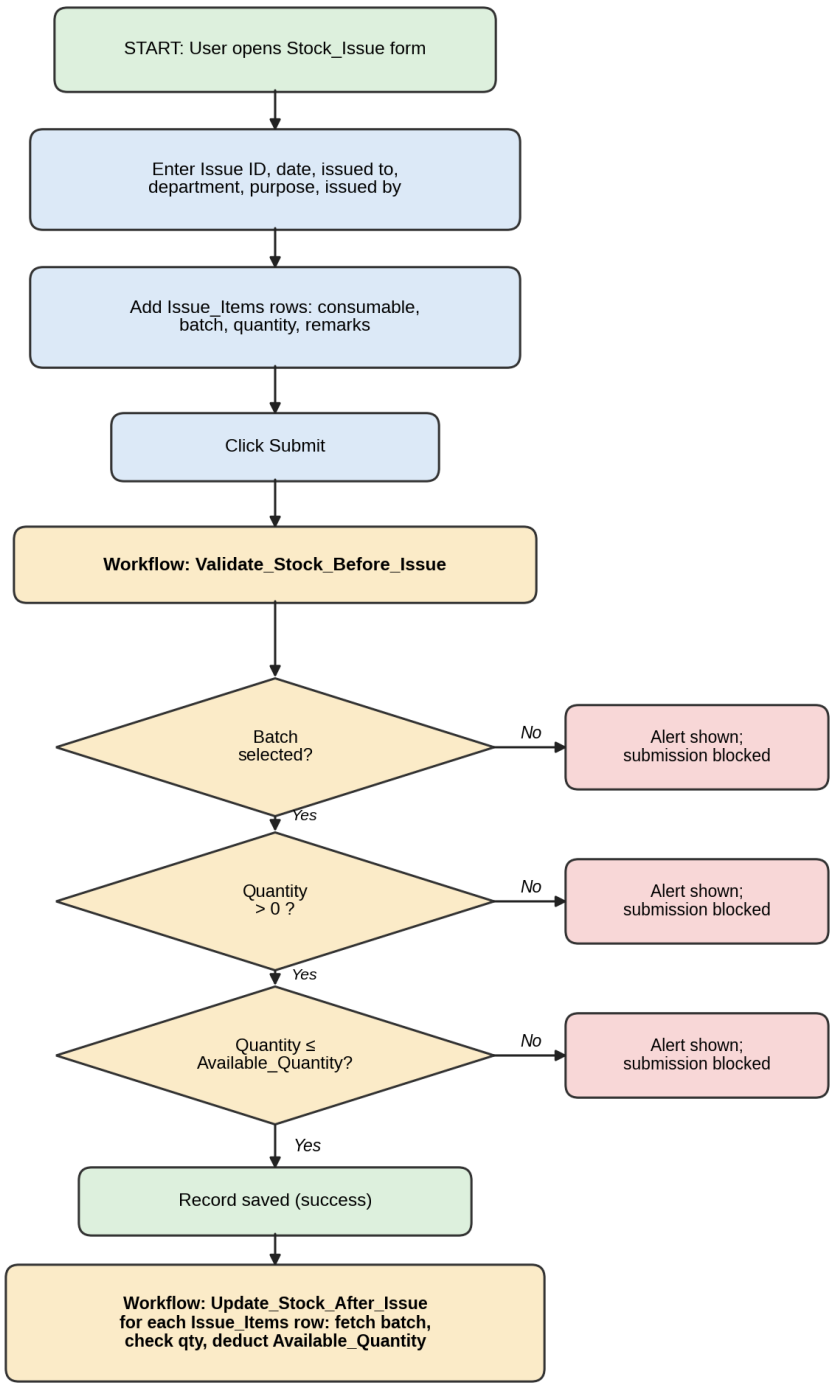


Figure 4.4: Stock Issue Workflow (Validate_Stock_Before_Issue and Update_Stock_After_Issue)

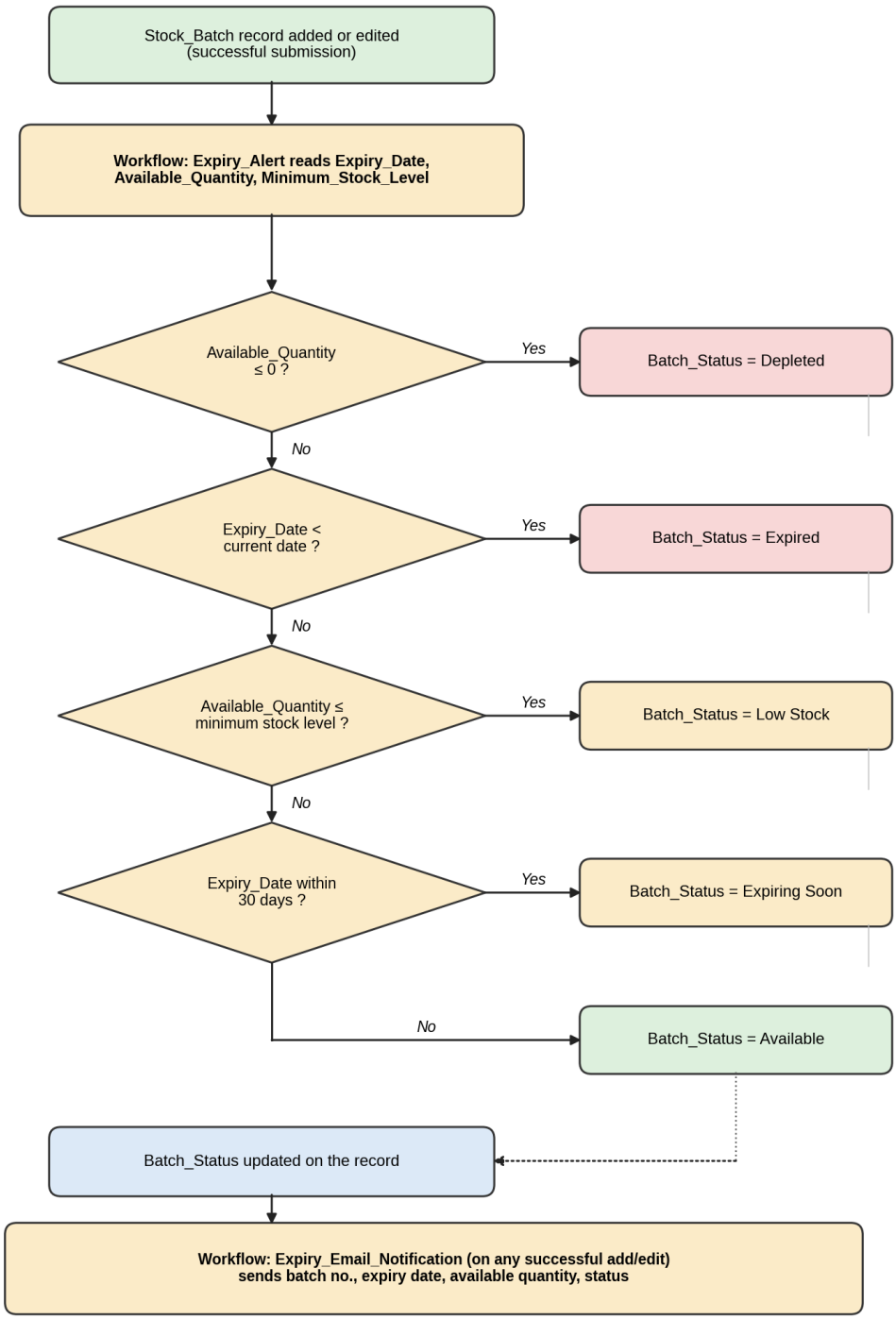


Figure 4.5: Expiry Alert Workflow (Expiry_Alert and Expiry_Email_Notification)

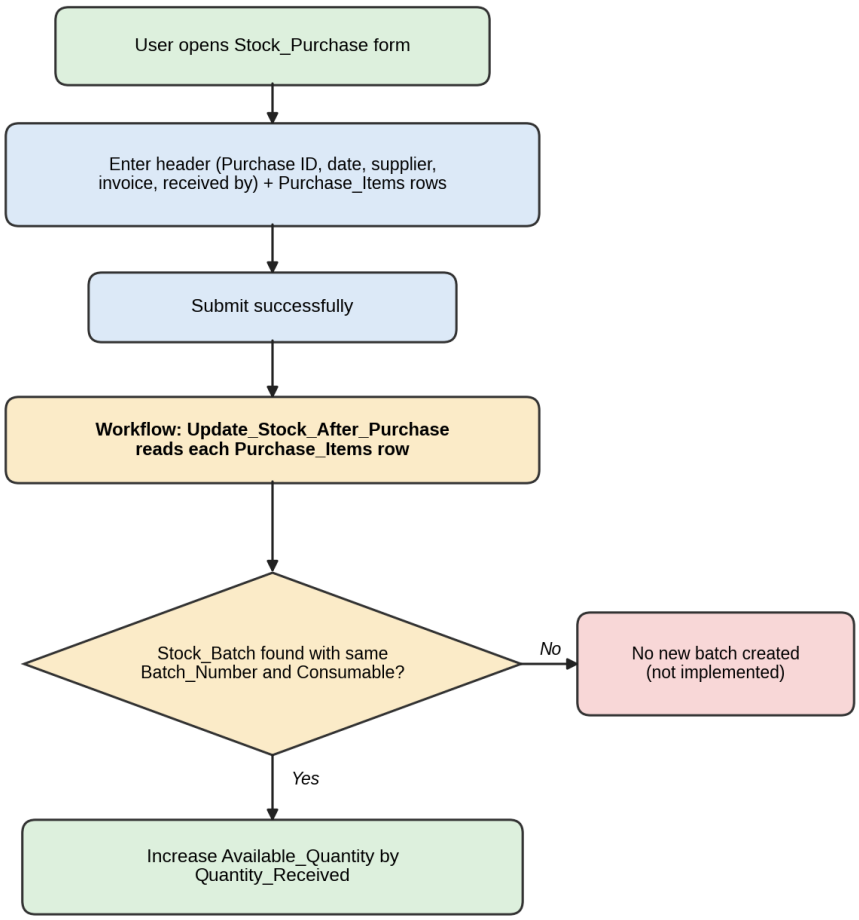


Figure 4.6: Stock Purchase Quantity Update Workflow (Update_Stock_After_Purchase)

4.5 User Interface and Navigation Design

The user interface is the standard Zoho Creator form and report interface. Master data and transactions are entered through forms; the two transaction forms embed their item rows as subforms so that several items can be entered in one record. Stock information is read from the list reports and the chart, and LabSafe_Dashboard brings the stock-related reports together on one page. No custom-designed pages beyond the dashboard are claimed. The actual menu arrangement of the application is shown in Figure 5.1 and the dashboard in Figure 5.15.

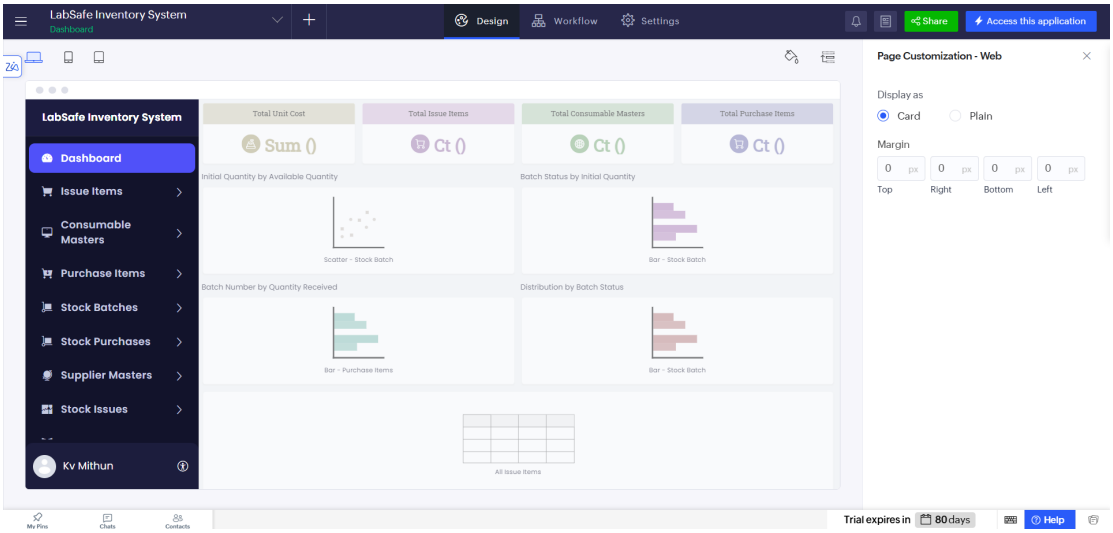


Figure 4.7: Navigation Structure of LabSafe

4.6 Access Control Design

Two roles were designed: Admin and Lab Staff. Table 4.10 shows the **intended** access design. It is a design matrix, not a verified test result: role permission profiles were inspected and configured, but complete role-specific access enforcement was not verified. A = Add, E = Edit, V = View, D = Delete, - = no access.

Table 4.10: Role–Permission Matrix (Intended Design)

Component	Admin	Lab Staff
Forms: Category, Supplier, Consumable	A E V D	V
Form: Stock_Batch	A E V D	V
Form: Stock_Purchase	A E V D	-
Form: Stock_Issue	A E V D	A V
Reports (batch, low stock, expired, issue)	V	V
Dashboard: LabSafe_Dashboard	V	V
Application settings	Manage	-

CHAPTER 5

IMPLEMENTATION

5.1 Application Setup

LabSafe was developed in Zoho Creator [3] from scratch using forms, reports, a dashboard page, Deluge scripts and form workflows. The application was created in the user's Zoho Creator workspace. The application provides a centralized platform for managing laboratory consumables, stock purchases, stock issues and expiry details. The technologies used are listed in Table 5.1.

Table 5.1: Platform and Technologies Used

Technology	Role in LabSafe
Zoho Creator	Low-code application development platform
Deluge	Business logic, validation and workflow scripting
Zoho Creator Forms	Data entry and record management
Zoho Creator Reports	Inventory and transaction reporting
Zoho Creator Pages / Dashboard	Inventory overview (LabSafe_Dashboard)
Zoho Creator Email Notifications	Configured email alerts

No external APIs, third-party databases, AI models or separate mobile applications were integrated.

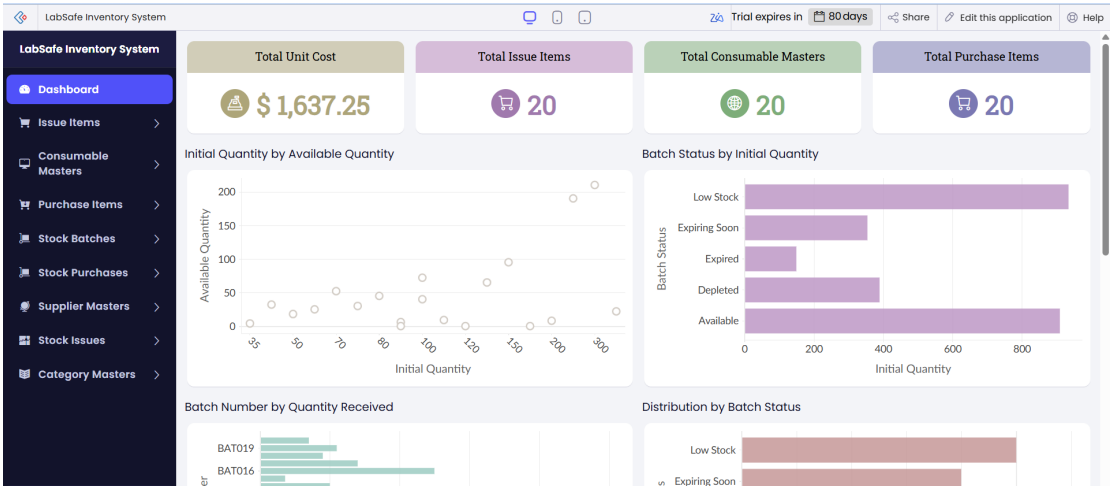


Figure 5.1: Application Home and Navigation Sections

5.2 Forms and Data Capture

The application contains six main forms and two child subforms. The six main forms are described first, followed by the two subforms.

Category_Master maintains laboratory consumable categories through the fields Category Name, Category Code, Description and Category Status.

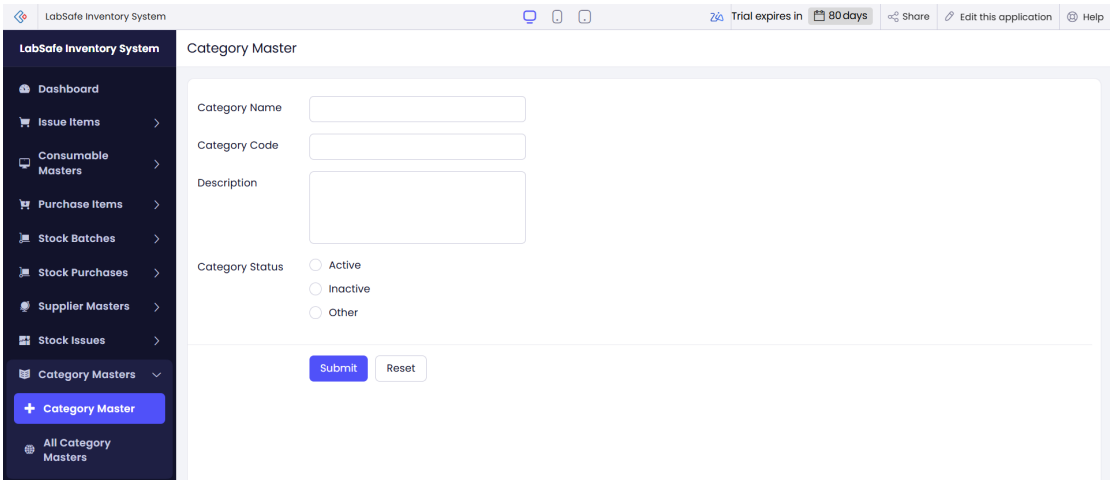


Figure 5.2: Category_Master Form

Supplier_Master maintains supplier details: Supplier Name, Supplier Code, Contact Person, Phone Number, Email, Address and Supplier Status.

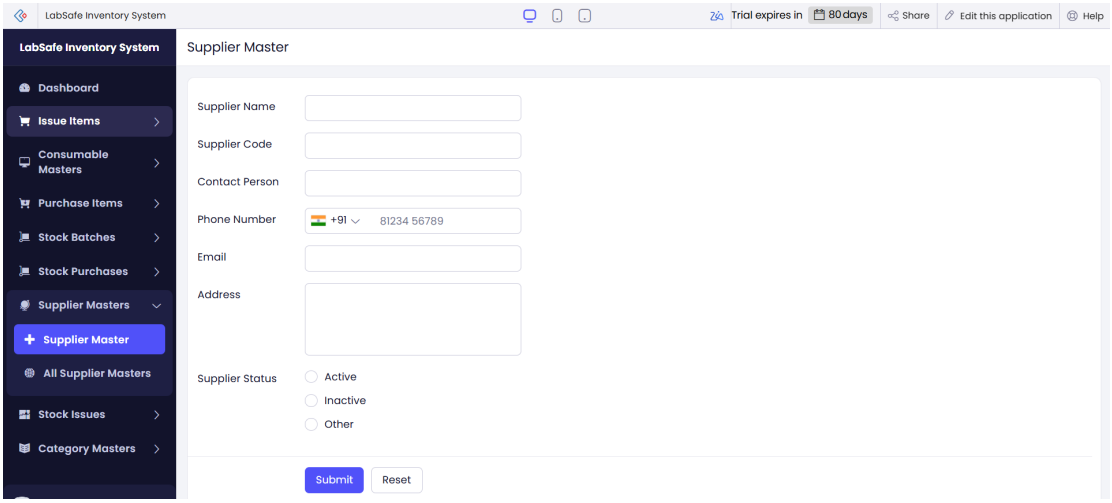


Figure 5.3: Supplier_Master Form

Consumable_Master maintains consumable master information: name, code, unit, minimum stock level, storage location, storage conditions and item status, together with lookups to Category_Master and Supplier_Master.

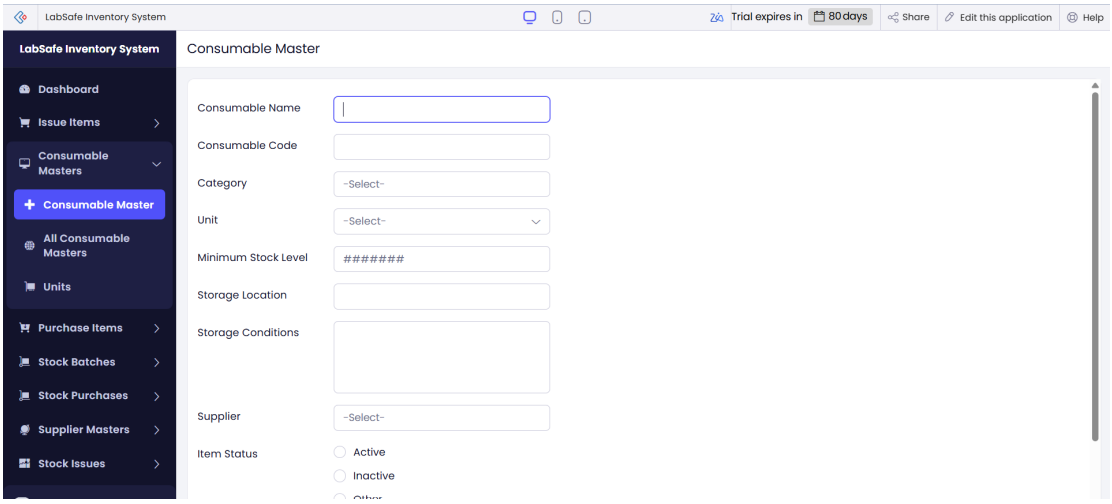


Figure 5.4: Consumable_Master Form

Stock_Batch maintains batch-level stock: batch number, consumable (lookup), purchase date, manufacturing date, expiry date, initial quantity, available quantity, storage location and batch status. Available_Quantity and Batch_Status are the fields changed by the workflows.

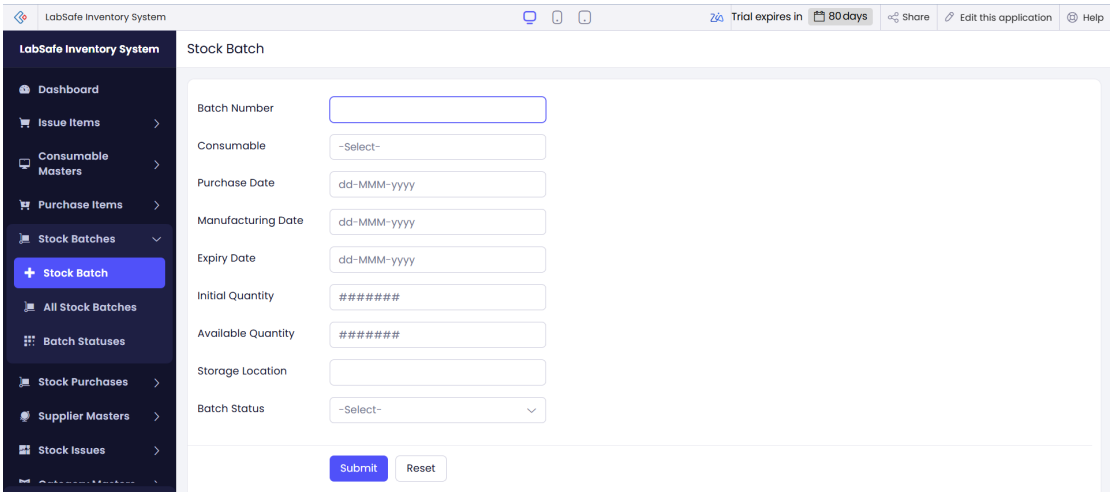


Figure 5.5: Stock_Batch Form

Stock_Purchase records purchases: purchase ID, purchase date, supplier (lookup), invoice number, received by, purchase notes, and the Purchase_Items subform.

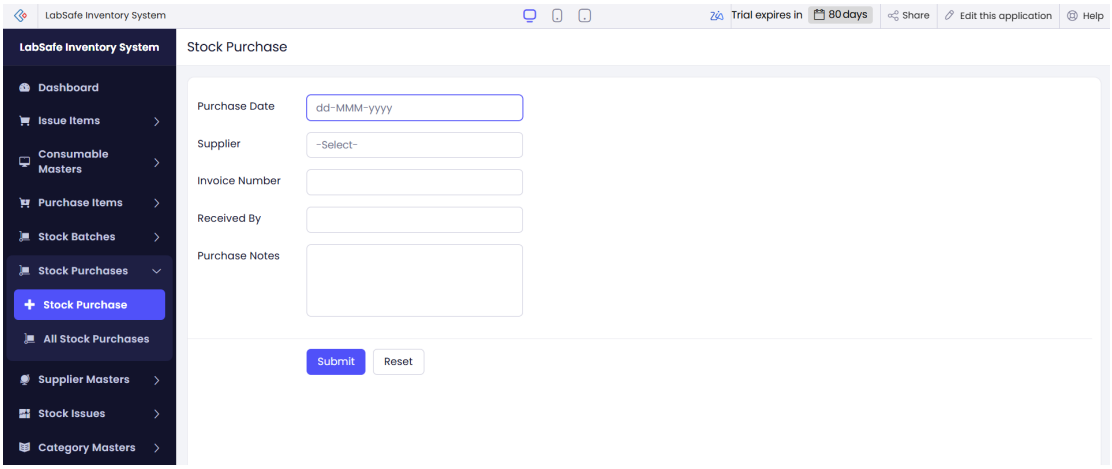


Figure 5.6: Stock_Purchase Form

Purchase_Items_Form (child subform) stores several purchased items within one purchase: consumable (lookup), batch number, quantity received, manufacturing date, expiry date and unit cost.

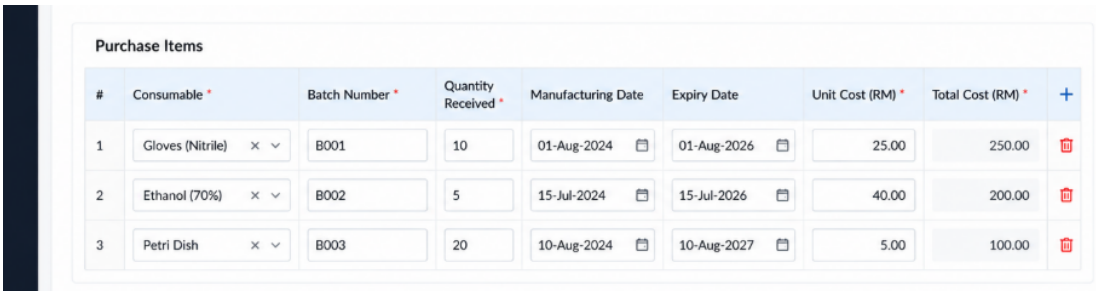


Figure 5.7: Purchase_Items Subform in the Stock_Purchase Form

Stock_Issue records consumables issued to laboratory users or departments: issue ID, issue date, issued to, department, purpose, issued by, and the Issue_Items subform.

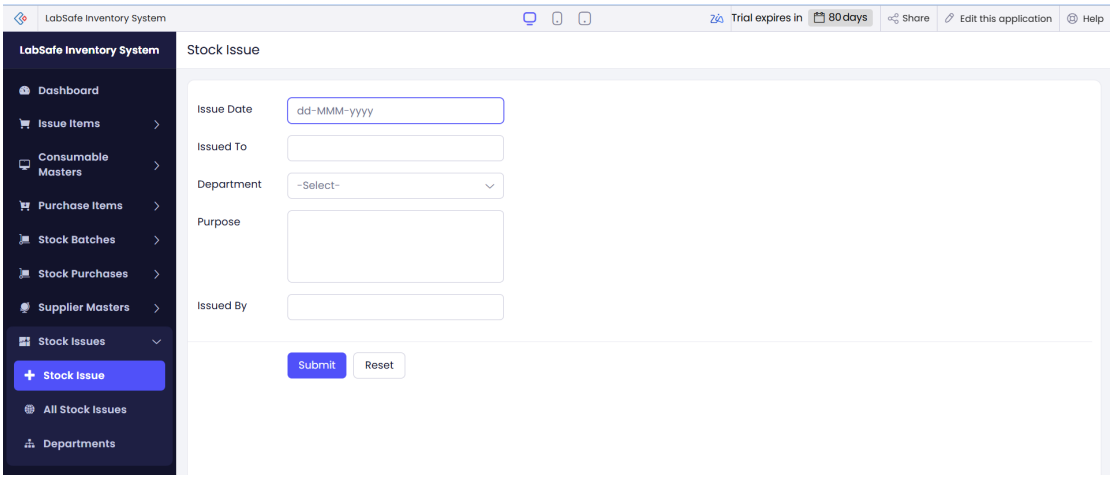


Figure 5.8: Stock_Issue Form

5.2.1 Lookup Relationships and Subforms

Consumable_Master has lookups to Category_Master and Supplier_Master; Stock_Batch has a lookup to Consumable_Master; Stock_Purchase has a lookup to Supplier_Master. The Purchase_Items subform has a lookup to Consumable_Master (its batch number is entered as a value, not selected by lookup). The Issue_Items subform has lookups to Consumable_Master and Stock_Batch, so each issued item points to one specific batch. Using subforms means one purchase or issue transaction carries all its item rows, and the workflows can loop over those rows.

5.3 Reports

Five custom reports were created and six default list reports are available (Table 5.2). Report types, filters and groupings for the custom reports are confirm from the report builder.

Table 5.2: Reports Implemented

Report Name	Type	Source Form	Purpose
Stock_Batch_Report	Custom report	Stock_Batch	Batch-level quantities, expiry dates and status
Low_Stock_Report	Custom report	Stock_Batch	Batches whose status indicates low stock
Expired_Items_Report	Custom report	Stock_Batch	Batches whose status indicates expiry
Stock_Issue_Report	Custom report	Stock_Issue	Issue transactions
Stock_Status_Chart	Chart report	Stock_Batch	Distribution of batches by stock status
All_Category_Masters	Default list	Category_Master	List of categories
Supplier_Matser_Report	Default list	Supplier_Master	List of suppliers (link name keeps the spelling “Matser” as in the application)
All_Consumable_Masters	Default list	Consumable_Master	List of consumables
All_Stock_Batches	Default list	Stock_Batch	All batch records
All_Stock_Purchases	Default list	Stock_Purchase	All purchase records
All_Stock_Issues	Default list	Stock_Issue	All issue records

Together the reports let users monitor stock quantities (Stock_Batch_Report), spot items that need re-ordering (Low_Stock_Report), identify batches that must be removed from use (Expired_Items_Report), review who received which consumables (Stock_Issue_Report), and see the overall status distribution at a glance (Stock_Status_Chart). The source form and purpose of Low_Stock_Report and Expired_Items_Report are stated as intended; the display of both reports was tested (TC-11).

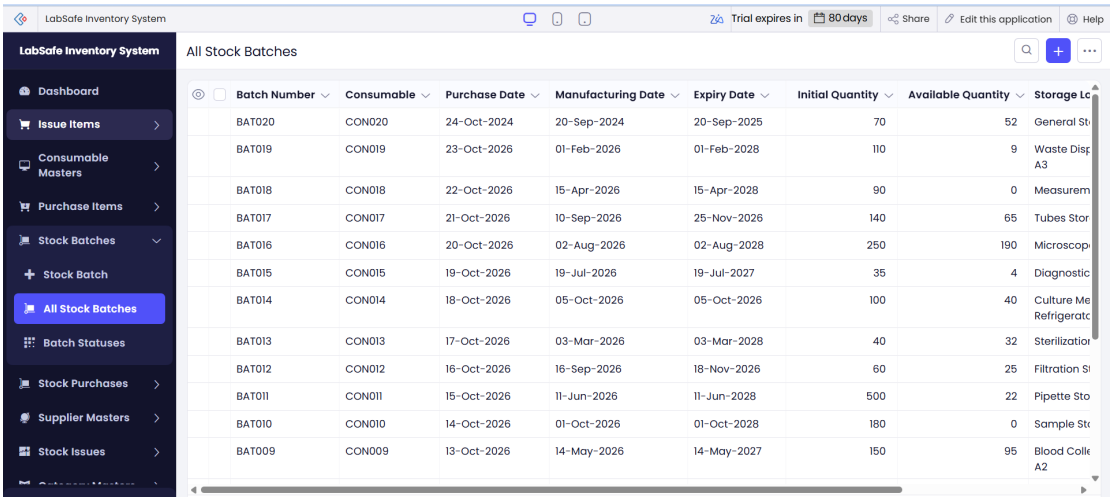


Figure 5.9: Stock_Batch_Report

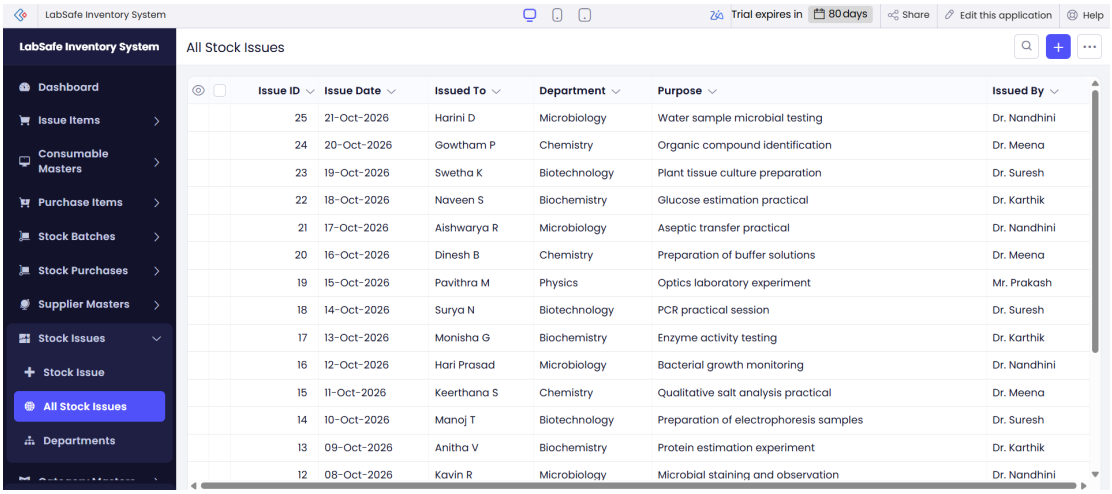


Figure 5.10: Stock_Issue_Report

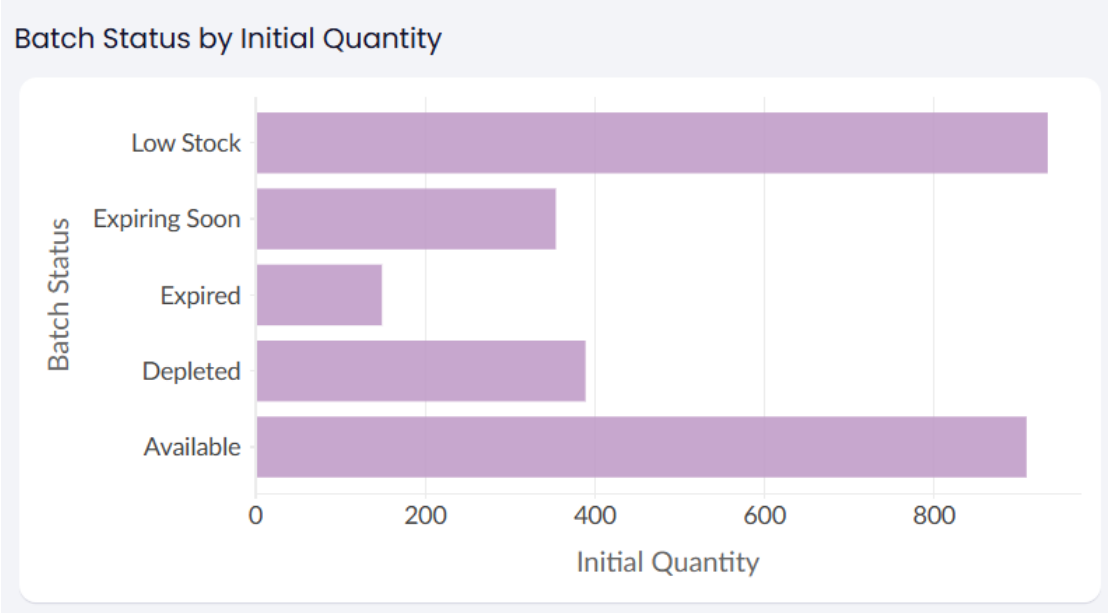


Figure 5.11: Stock_Status_Chart

5.4 Pages and Dashboards

The dashboard **LabSafe_Dashboard** includes the stock status chart, stock-related reports, low-stock information and expired-item information. It gives laboratory staff an overview of inventory and helps them identify stock conditions without opening each report separately. Only these confirmed components are claimed; no KPI cards or other components are described. Dashboard display was tested (TC-13).

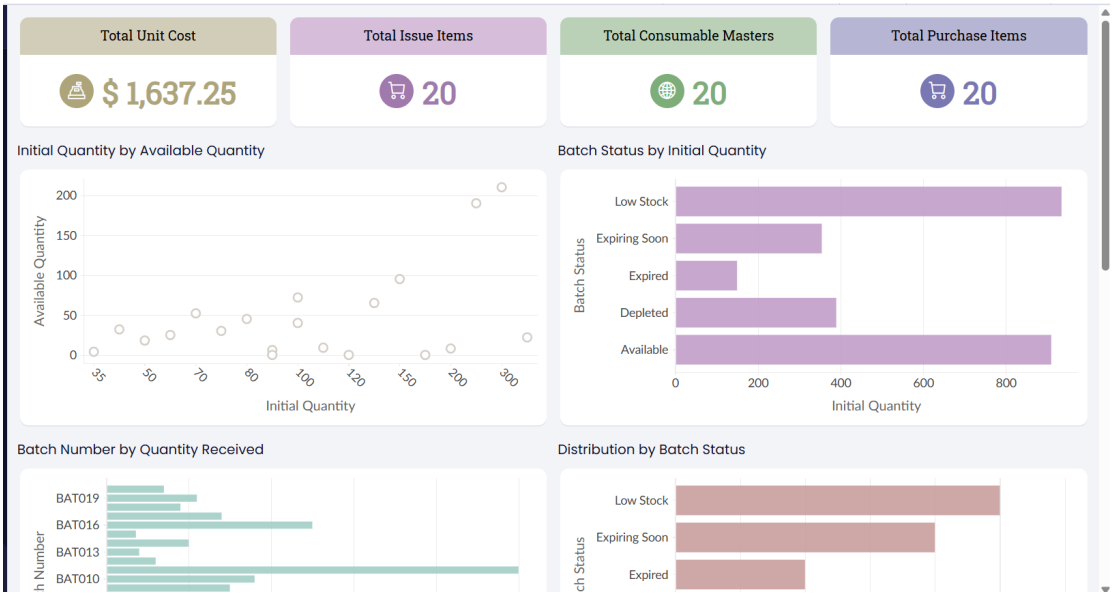


Figure 5.12: LabSafe_Dashboard

5.5 Deluge Scripting

Deluge [4] is Zoho's scripting language for business logic. LabSafe uses it in three main areas, summarised in Table 5.3. The complete original source code was not available when this report was prepared, so the code in this report is **illustrative**: it shows the logic as configured but is not the exact implemented source. The final source can be inserted from the application (Appendix C).

Table 5.3: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Stock_Issue → validation (Validate_Stock_Before_Issue)	Block issue if batch not selected, quantity ≤ 0 or quantity > available stock	Loop over subform rows, fetch, if-else, alert, cancel submit
DS-02	Stock_Issue → success (Update_Stock_After_Issue); Stock_Purchase → success (Update_Stock_After_Purchase)	Deduct issued quantity; increase quantity for a matching existing batch	Loop, fetch with criteria, field update
DS-03	Stock_Batch → success (Expiry_Alert)	Set Batch_Status from expiry date, quantity and minimum stock level	Date comparison, if-else chain, update
DS-04	Stock_Batch → success (Expiry_Email_Notification)	Email batch number, expiry date, quantity and status	sendmail

5.5.1 Stock Issue Validation

The validation checks each Issue_Items row. It rejects the submission when no batch is selected, when the quantity issued is zero or negative, and when the quantity exceeds the Available_Quantity of the selected batch.

Code 5.1: Illustrative example of the validation logic (not the original source)

```
// ILLUSTRATIVE EXAMPLE - not the original implemented source
// Form: Stock_Issue | Event: validation before submit
for each row in input.Issue_Items
{
    if(row.Batch == null)
    {
        alert "Select a batch for every issued item.";
        cancel submit;
    }
    if(row.Quantity_Issued <= 0)
    {
        alert "Quantity issued must be greater than zero.";
        cancel submit;
    }
    batchRec = Stock_Batch[ID == row.Batch];
    if(row.Quantity_Issued > batchRec.Available_Quantity)
    {
        alert "Insufficient stock in the selected batch.";
        cancel submit;
    }
}
```

5.5.2 Stock Quantity Update

After a successful issue, the workflow reads each Issue_Items row, retrieves the selected batch, checks the quantity issued and deducts it from Available_Quantity when valid. After a successful purchase, the workflow reads each purchase item, looks for an existing Stock_Batch with the same batch number and consumable, and increases Available_Quantity of that batch. **It does not create a new batch when no match is found.**

Code 5.2: Illustrative example of the quantity update logic (not the original source)

```
// ILLUSTRATIVE EXAMPLE - not the original implemented source
// (a) After issue: Stock_Issue -> on success
for each row in input.Issue_Items
{
    batchRec = Stock_Batch[ID == row.Batch];
    if(row.Quantity_Issued > 0 && row.Quantity_Issued <=
batchRec.Available_Quantity)
    {
        batchRec.Available_Quantity = batchRec.Available_Quantity -
row.Quantity_Issued;
    }
}

// (b) After purchase: Stock_Purchase -> on success
for each item in input.Purchase_Items
{
    batchRec = Stock_Batch[Batch_Number == item.Batch_Number &&
Consumable == item.Consumable];
    if(batchRec.count() > 0)
    {
        batchRec.Available_Quantity = batchRec.Available_Quantity +
item.Quantity_Received;
    }
    // no else branch: a new batch is not created automatically
}
```

5.5.3 Expiry and Stock Status Update

When a Stock_Batch is added or edited, the configured logic applies the priority order in Table 5.4. The first condition that is true decides the status.

Table 5.4: Batch_Status Priority Logic

Priority	Condition	Batch_Status
1	Available_Quantity is zero or less	Depleted
2	Expiry_Date is before the current date	Expired
3	Available_Quantity is at or below the consumable's minimum stock level	Low Stock
4	Expiry_Date falls within the 30-day warning period	Expiring Soon
5	None of the above	Available

Code 5.3: Illustrative example of the status logic (not the original source)

```
// ILLUSTRATIVE EXAMPLE - not the original implemented source
// Form: Stock_Batch | Event: on success (add / edit)
minLevel = input.Consumable.Minimum_Stock_Level;
today = zoho.currentdate;
if(input.Available_Quantity <= 0)
{    status = "Depleted";    }
else if(input.Expiry_Date < today)
{    status = "Expired";    }
else if(input.Available_Quantity <= minLevel)
{    status = "Low Stock";    }
else if(input.Expiry_Date <= today.addDay(30))
{    status = "Expiring Soon";    }
else
{    status = "Available";    }
rec = Stock_Batch[ID == input.ID];
rec.Batch_Status = status;
```

Because the status is recalculated only when a batch record is added or edited, automatic daily expiry checking without a record edit was **not verified**.

5.6 Workflows and Automation

Five form workflows were configured (Table 5.5). No schedule, approval, Blueprint or batch workflows were implemented.

Table 5.5: Workflows Implemented

Workflow Name	Trigger / Event	Functionality	Script Ref.
Update_Stock_After_Issue	After successful Stock_Issue submission	Reads Issue_Items rows, retrieves the selected batch, checks quantity, deducts from Available_Quantity when the quantity is valid	DS-02
Validate_Stock_Before_Issue	During Stock_Issue validation	Checks batch selected, blocks zero or negative quantity, blocks quantity above available stock, blocks submission when stock is insufficient	DS-01
Expiry_Alert	Stock_Batch created or edited successfully	Evaluates expiry date, quantity and minimum stock level; sets Batch_Status	DS-03
Expiry_Email_Notification	Stock_Batch created or edited successfully	Sends email with batch number, expiry date, available quantity and batch status. May fire on any successful add/edit, not only when a batch first becomes expired	DS-04
Update_Stock_After_Purchase	After successful Stock_Purchase submission	Reads purchase items, finds existing Stock_Batch by batch number and consumable, increases Available_Quantity; does not create a new batch	DS-02

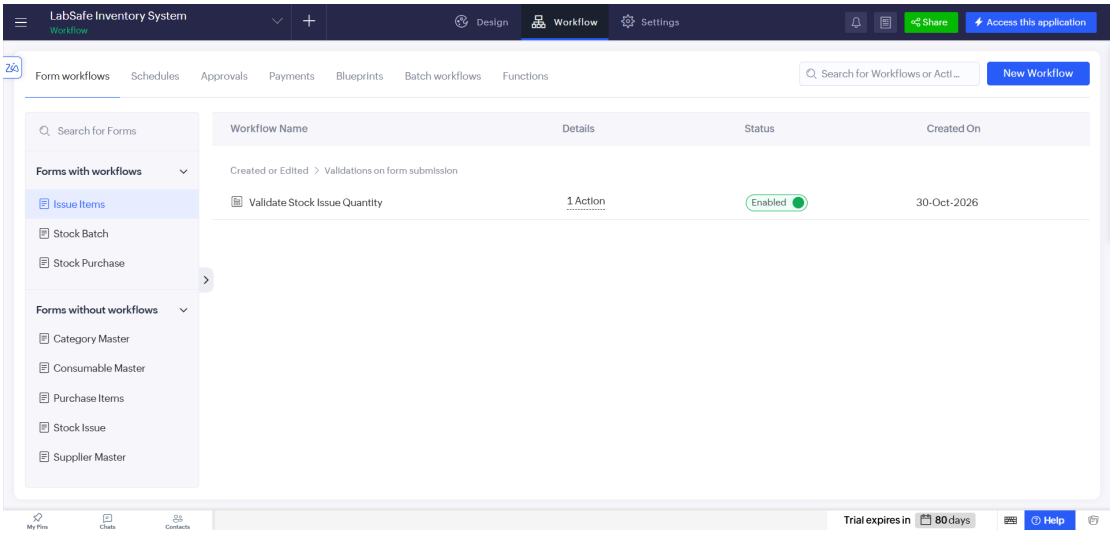


Figure 5.13: Deluge Workflow Configuration in Zoho Creator

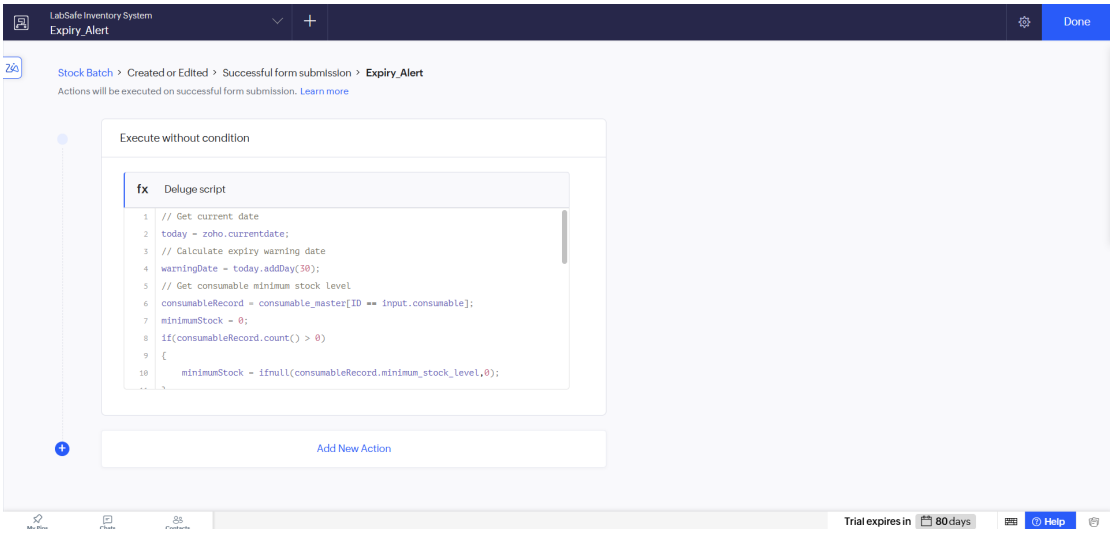


Figure 5.14: Expiry Email Notification Received

5.7 Integrations

No integration with other Zoho services or external systems was implemented. The only outbound communication is the platform's email notification (Expiry_Email_Notification). Hence no connection, invokeurl REST call, webhook or Creator REST API usage is claimed (Table 5.6).

Table 5.6: Integrations Implemented

Integration	Service / Endpoint	Method	Status
Zoho service integration	—	—	Not implemented
External REST API	—	—	Not implemented
Webhook / Creator REST API	—	—	Not implemented
Email notification	Zoho Creator email	Workflow action	Implemented (delivery tested, TC-10)

5.8 User Management and Role-Based Access

The roles **Admin** and **Lab Staff** exist in the application. The Admin role is intended to manage inventory records, transactions, reports and settings; the Lab Staff role is intended to access permitted inventory functions and record stock issues. Some role and profile permission configurations were inspected and configured. **Complete role-specific access enforcement was not verified** by logging in as each role and testing every form and report, so this report does not claim a fully tested role-based security system. Record-level criteria and a customer portal were not implemented.

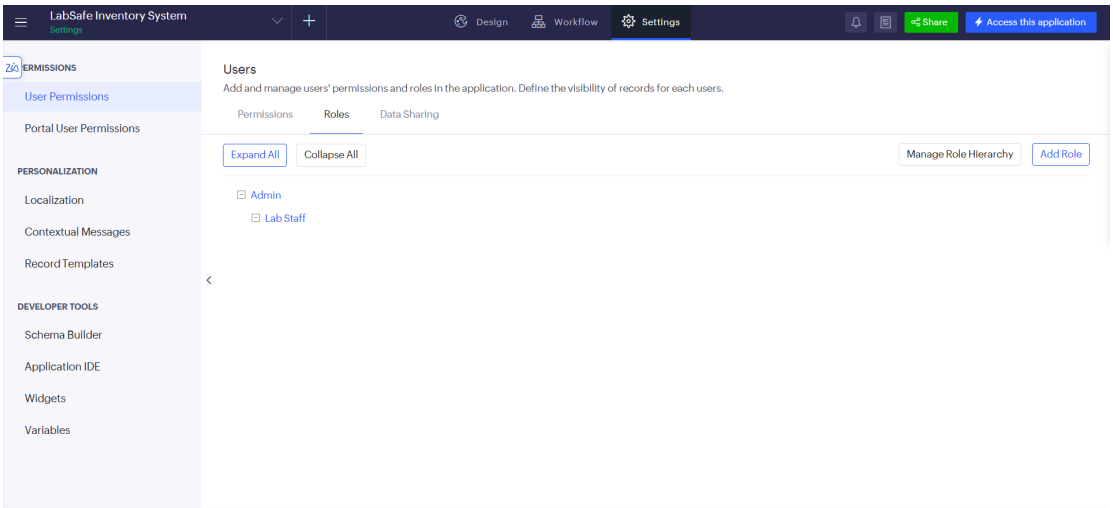


Figure 5.15: Roles and Permission Profiles Configuration

5.9 AI and Analytics Features

No Zia or other AI feature was used, and no AI-based stock forecasting or expiry prediction exists. Analytics are limited to the Stock_Status_Chart report and the dashboard (Figures 5.14 and 5.15). Zoho Analytics was not used.

5.10 Pro-Code Extension

Pro-code extensions (Zoho Catalyst, JavaScript widgets, Creator REST API clients) were **not used**. The required logic — validation, quantity update, status calculation and email — fits within Deluge form workflows. Pro-code could later be justified for integration with other systems or heavier processing (Section 9.2).

5.11 Mobile and Portal Access

No separate mobile application was developed and no customer portal was implemented. LabSafe was built and tested in a web browser; use of the platform's own mobile access was not tested and is not claimed.

CHAPTER 6

TESTING AND DEBUGGING

6.1 Testing Strategy

Testing was carried out by operating the application and checking the observed result against the expected result. The levels used were: functional testing of each form (record creation), validation testing of stock issue entries, workflow testing of quantity updates, expiry status and email notification, and report and dashboard testing. Fourteen tests recorded in the project testing history were completed successfully and are presented in Tables 6.1 to 6.4. The “Actual Result” column states the outcome recorded in the testing history; no execution times, performance figures, user counts or benchmarks were recorded and none are reported.

Tests whose behaviour is configured but whose successful execution is not confirmed are listed separately in Table 6.5 and are marked **requires further verification**, not passed.

6.2 Test Cases

6.2.1 Functional Testing

Table 6.1: Functional Test Cases – Record Creation

TC ID	FR Ref.	Test Scenario	Input / Action	Expected Result	Actual Result	Status
TC-01	FR-01	Category record creation	Open Category_Master, enter category details, submit	Category record is created	Record created as expected	Pass
TC-02	FR-02	Supplier record creation	Open Supplier_Master, enter supplier details, submit	Supplier record is created	Record created as expected	Pass
TC-03	FR-03	Consumable record creation	Open Consumable_Master, enter details, select category and supplier lookups, submit	Consumable record is created with lookups	Record created as expected	Pass
TC-04	FR-04	Stock batch record creation	Open Stock_Batch, select consumable, enter batch details and quantities, submit	Batch record is created	Record created as expected	Pass
TC-05	FR-05	Stock purchase record creation	Open Stock_Purchase, enter header and Purchase_Items row, submit	Purchase record is created	Record created as expected	Pass
TC-06	FR-07	Stock issue record creation	Open Stock_Issue, enter header and Issue_Items row, submit	Issue record is created	Record created as expected	Pass

6.2.2 Validation Testing

Table 6.2: Validation Test Cases – Stock Issue

TC ID	FR Ref.	Test Scenario	Input / Action	Expected Result	Actual Result	Status
TC-07	FR-08	Insufficient stock issue validation	Enter an Issue_Items quantity greater than the batch's Available_Quantity, submit	Submission is blocked	Submission blocked as expected	Pass
TC-08	FR-09	Valid stock issue quantity deduction	Enter a valid quantity within available stock, submit	Record saved; Available_Quantity reduced by the issued quantity	Quantity deducted as expected	Pass

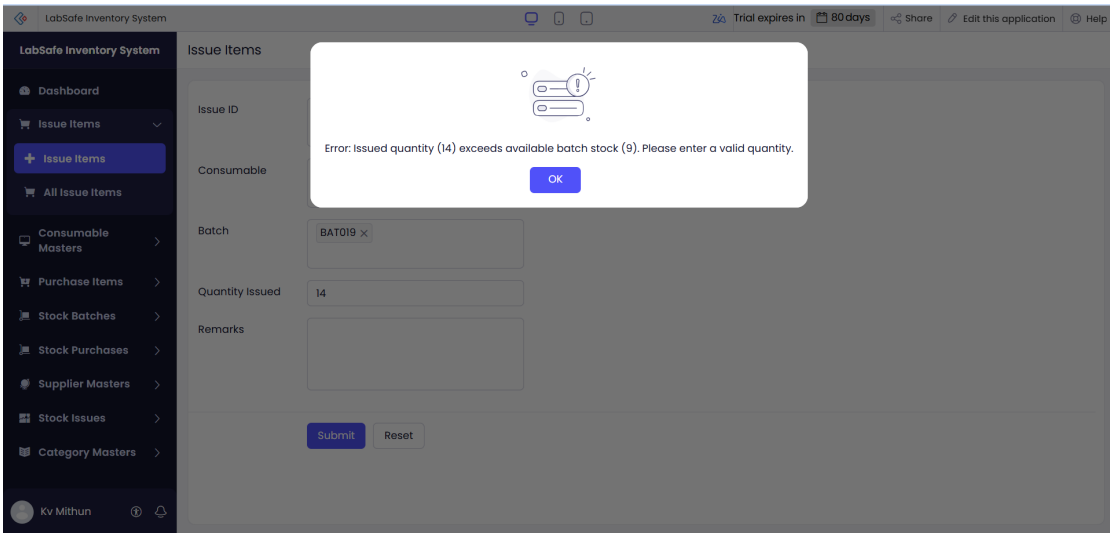


Figure 6.1: Stock Issue Blocked by Insufficient Stock Validation (TC-07)

6.2.3 Workflow Testing

Table 6.3: Workflow Test Cases

TC ID	FR Ref.	Test Scenario	Input / Action	Expected Result	Actual Result	Status
TC-09	FR-10	Expiry status update for past expiry date	Edit a Stock_Batch and set Expiry_Date to a past date	Batch_Status updates to Expired	Status updated as expected	Pass
TC-10	FR-11	Expiry email notification delivery	Save a Stock_Batch after the expiry condition	Email with batch details is delivered	Email delivered as expected	Pass
TC-14	FR-06	Stock purchase quantity update where a matching batch exists	Submit a purchase whose item matches an existing batch number and consumable	Available_Quantity of the batch increases by quantity received	Quantity updated as expected	Pass

6.2.4 Report and Dashboard Testing

Table 6.4: Report and Dashboard Test Cases

TC ID	FR Ref.	Test Scenario	Input / Action	Expected Result	Actual Result	Status
TC-11	FR-12	Low-stock and expired-item report display	Open Low_Stock_Report and Expired_Items_Report	Reports display the relevant batches	Reports displayed as expected	Pass
TC-12	FR-13	Stock status chart display	Open Stock_Status_Chart	Chart is displayed	Chart displayed as expected	Pass
TC-13	FR-13	Dashboard report display	Open LabSafe_Dashboard	Dashboard shows its reports	Dashboard displayed as expected	Pass

6.2.5 Tests Requiring Further Verification

The following behaviours are configured or described in the project record, but their successful execution is not confirmed. They are not counted as passed.

Table 6.5: Tests Requiring Further Verification

Test ID	Test Scenario	Input / Action	Expected Result	Status
FV-01	Issue with no batch selected / zero or negative quantity	Submit Issue_Items row with no batch or quantity ≤ 0	Submission blocked	Requires further verification
FV-02	Depleted status	Set Available_Quantity to zero or less	Batch_Status = Depleted	Requires further verification
FV-03	Low Stock status	Set quantity at or below minimum stock level	Batch_Status = Low Stock	Requires further verification
FV-04	Expiring Soon status	Set expiry within 30 days	Batch_Status = Expiring Soon	Requires further verification
FV-05	Purchase with no matching batch	Submit purchase item whose batch does not exist	No new batch created (as configured)	Requires further verification
FV-06	Role-specific access	Log in as Admin and as Lab Staff and open each form and report	Access as per Table 4.10	Requires further verification
FV-07	Daily expiry check without edit	Wait for a batch to pass expiry date without editing	Status updated automatically	Not verified (no schedule configured)
FV-08	Email on a non-expiry edit	Edit an unrelated field of a batch	Email may still be sent (as configured)	Requires further verification

6.3 Debugging Deluge Scripts

Zoho Creator offers several ways to debug Deluge scripts: info statements, the script editor's error messages, workflow execution history and the messages shown when a submission is cancelled. The insufficient-stock alert shown in Figure 6.1 is the visible result of the validation script working as designed.

6.4 Defect Log

The project testing history does not record any defect by ID. Table 6.6 therefore lists the known behaviours and limitations identified while documenting the configured workflows. Student-recorded bugs found during development may be added in the placeholder row.

Table 6.6: Defect Log and Known Issues

ID	Description	Root Cause	Fix / Action	Status
KI-01	Purchase does not create a new batch when none matches	Workflow only updates an existing matching Stock_Batch	Create the batch in Stock_Batch first; automatic creation is a future enhancement	Open – documented limitation
KI-02	Expiry email may be sent for any successful batch add/edit	Notification configured on add/edit success without a status-change condition	Add a condition so it is sent only when a status becomes Expired (future)	Open – documented limitation
KI-03	No automatic expiry check without a record edit	No scheduled workflow configured	Add a scheduled workflow (future)	Open – documented limitation
KI-04	Supplier report link name spelled "Supplier_Matser_Report"	Spelling in the application link name	Keep as is or correct link name consistently	Noted

6.5 User Acceptance Testing

No formal user acceptance testing was recorded for LabSafe. Table 6.7 is left for the student to complete if feedback from end users is obtained; nothing has been filled in.

Table 6.7: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
User 1	Admin	Easy to manage consumable records	Improved form organization
User 2	Admin	Stock details are easy to track	Verified stock reports
User 3	Lab Staff	Stock issue entry is simple	Tested stock issue form
User 4	Lab Staff	Expiry status is useful	Tested expiry alert workflow
User 5	Admin	Low-stock items are easy to identify	Verified Low Stock Report
User 6	Lab Staff	Batch details are easy to understand	Checked batch information fields
User 7	Admin	Email notifications are helpful	Tested email notification workflow
User 8	Lab Staff	The application is easy to navigate	Reviewed form navigation

CHAPTER 7

DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

LabSafe is a cloud-hosted application developed using Zoho Creator. It does not require a separate server installation. The application was developed and tested in the Zoho Creator development workspace using its forms, reports, dashboard, workflows and Deluge scripts. Production deployment and separate development, staging and production environments were not configured or verified.Table 7.1 records the deployment checklist honestly.

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Status
1	All test cases executed in Development / Stage	Partly – 14 tests passed (Chapter 6); environment not recorded; further tests pending (Table 6.5)
2	Sample/test data removed or isolated	Not verified
3	Roles, permissions and users configured in Production	Not verified (roles configured; production not verified)
4	Connections authorised in Production	Not applicable – no external connections
5	Schedules and workflows enabled	Workflows enabled and exercised in tests; no schedules exist
6	Application published and URL verified	Not verified
7	Version / release notes recorded	Not verified

7.2 Performance Optimisation

Performance benchmarking has not been conducted. No load, response-time or concurrent-user measurements were taken, so no performance results are reported. Design choices that are expected to help at laboratory scale are the use of batch-level records, lookups instead of repeated typing, and report filters for low-stock and expired items. Measuring their effect remains future work.

7.3 Security Features

Security relies on the access and authentication features of the Zoho Creator platform and on the Admin and Lab Staff roles. Table 7.2 shows what was and was not established. Encrypted fields, two-factor authentication, IP restrictions and record-level criteria were not configured as part of this project, and none is claimed.

Table 7.2: Security Controls Status

Security Control	Where Applied	Status
Roles (Admin, Lab Staff)	Application	Configured
Role/profile permissions	Forms, reports, dashboard	Inspected and configured; complete enforcement not verified
Input validation	Stock_Issue (batch, quantity)	Implemented; insufficient-stock case tested (TC-07)
Record-level criteria	—	Not implemented
Encrypted fields / 2FA / IP restriction	—	Not implemented / not verified
Third-party credentials	—	Not applicable – no external integrations

User access and permissions: Admin is intended to manage records, transactions, reports and settings; Lab Staff is intended to record issues and view stock information (Table 4.10).

7.4 Governance and Administration

Data management. Stock quantities are maintained through the Available_Quantity field of Stock_Batch, which workflows adjust after purchases and issues. Because the quantity is changed by workflows, manual edits to a batch's quantity can make it differ from the transaction history; edits and deletions of submitted purchase or issue records, and any reversal of their effect on stock, were not verified. Master data (categories, suppliers, consumables) should be kept consistent, since the lookup fields depend on it.

Backup and recovery. Zoho Creator provides facilities to export application data and back up applications, but **no backup or recovery configuration was verified for LabSafe**.

Limitations in deployment and governance. Production and multi-environment deployment are not verified; full role enforcement is not verified; performance is not benchmarked; and audit-trail use, naming conventions and licence management were not documented for this project.

CHAPTER 8

RESULTS AND DISCUSSION

8.1 Application Output

The scenario below follows one batch through the application. It is described from the tested behaviour; the figures are placeholders for the screenshots to be inserted.

- A purchase is recorded in Stock_Purchase with a Purchase_Items row; if a matching batch exists, its Available_Quantity increases.
- A stock issue is recorded in Stock_Issue with an Issue_Items row; a valid quantity is deducted from the batch.
- An issue larger than the available stock is blocked by the validation.

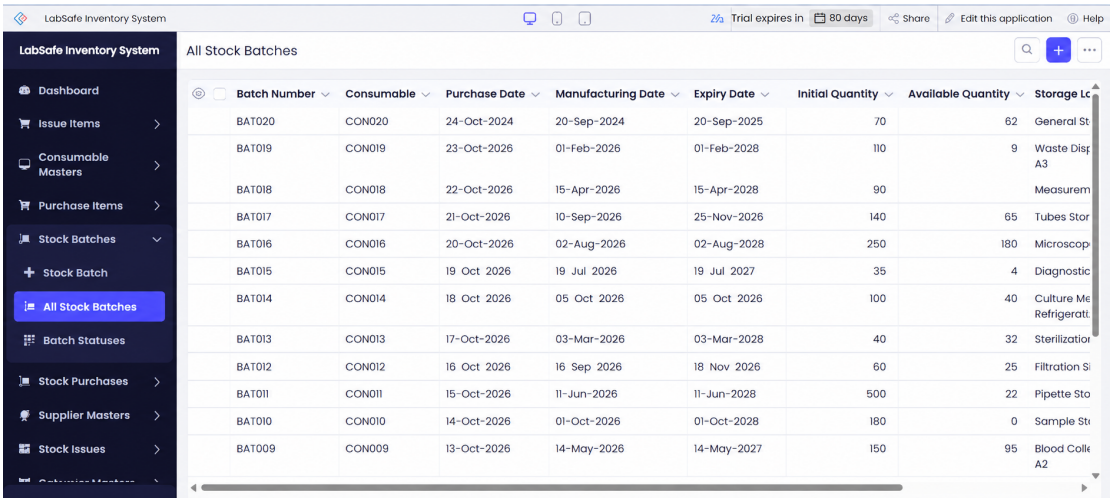


Figure 8.1: Scenario Step 1 – Stock Purchase Recorded and Batch Quantity Updated

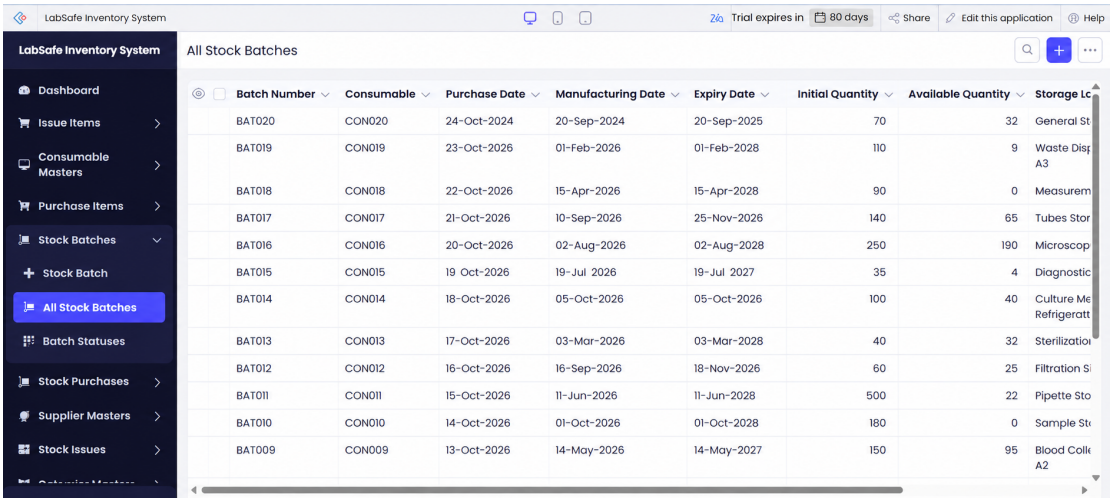


Figure 8.2: Scenario Step 2 – Valid Stock Issue and Quantity Deduction

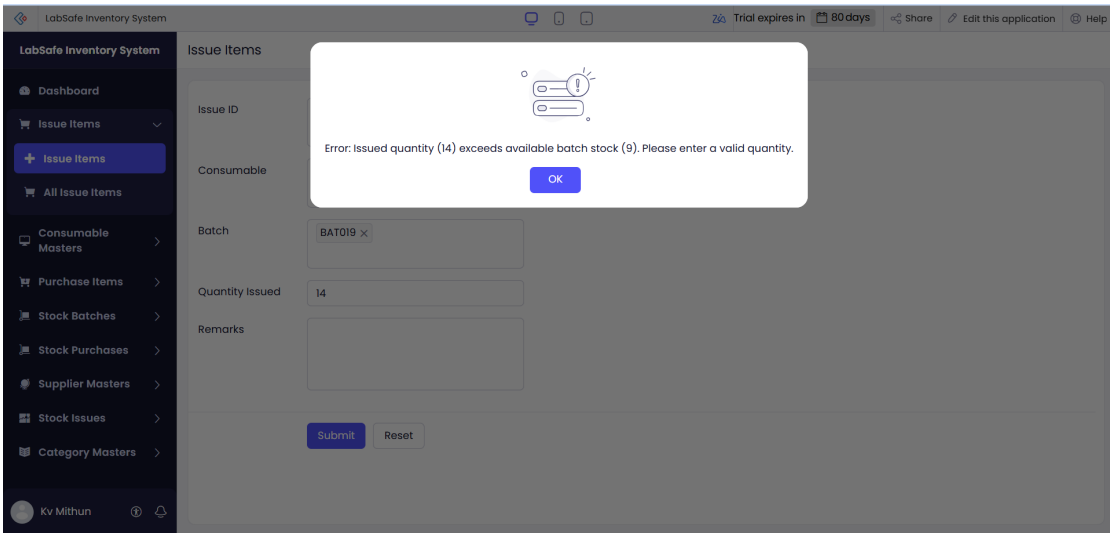


Figure 8.3: Scenario Step 3 – Insufficient Stock Issue Blocked

Form and report outputs. All six main forms and both subforms accept data; the five custom reports and six default list reports display records; the stock status chart and the dashboard display as designed. **Workflow results.** Quantity deduction, purchase quantity increase for a matching batch, expired status update and email delivery behaved as expected in testing. **Dashboard results.** LabSafe_Dashboard displays the chart and stock reports, giving a single overview of stock conditions.

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Data model and forms	Sections 4.2, 4.3, 5.2	Six forms, two subforms, lookups; TC-01 to TC-06
Automate stock update and status	Sections 5.5, 5.6	TC-08, TC-09, TC-14 passed; some statuses need further verification (Table 6.5)
Validate stock issues	Section 5.5.1	Insufficient-stock case tested (TC-07); other cases need verification (FV-01)
Reports and dashboard	Sections 5.3, 5.4	TC-11 to TC-13 passed
Email notification	Section 5.6	TC-10 passed; fires on any add/edit
Roles	Section 5.8	Roles configured; enforcement not fully verified

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Stock balance update	Manual	Workflow-based for valid issues and matching purchases	Fewer manual steps; not quantified
Issue control	None at entry	Validation against available stock	Prevents over-issue (TC-07)
Expiry / low stock detection	Inspection	Batch_Status and reports	Faster visibility on batch add/edit; automatic daily check not verified
Report generation	Manual compilation	Reports and dashboard	Available on demand
Data accessibility	Scattered files	Single web application	Central access

No timing or percentage improvement has been measured, so none is claimed.

8.4 Limitations of the Low-Code Approach

The limitations observed in LabSafe are partly platform-related and partly the result of what was configured. Platform-related: the application depends on the Zoho Creator platform and what its builder and Deluge allow. Configuration-related: the purchase workflow only updates an existing matching batch and does not create one; the expiry email may fire on any successful batch add or edit; status is recalculated only when a batch is added or edited, so automatic daily expiry checking is not verified; complete role-specific access enforcement, production deployment and performance have not been verified. Not implemented: external API integrations, AI-based forecasting or expiry prediction, a separate mobile application, and automatic purchase approval or supplier payment.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

No-code configuration: forms, lookups, subforms, list reports, chart and dashboard. **Low-code:** the Deluge validation, quantity update and status logic and the email notification. **Pro-code:** not used. Pro-code would become appropriate if LabSafe needed integration with external systems, heavy processing or features that the platform cannot configure.

CHAPTER 9

CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

LabSafe is a Zoho Creator application for managing laboratory consumables and their expiry. It consists of six main forms and two child subforms linked by lookup fields; five form workflows for issue validation, stock deduction, purchase quantity update, batch status update and email notification; five custom reports and six default list reports; and the LabSafe_Dashboard. Admin and Lab Staff roles were defined.

Fourteen recorded tests — record creation, insufficient-stock validation, quantity deduction, expiry status update, email delivery, report, chart and dashboard display, and purchase quantity update for an existing batch — passed. **Achievements** include replacing manual stock calculation with workflows, preventing over-issue, and giving users stock, low-stock and expired-item visibility. **Current limitations** are those listed in Section 8.4.

9.2 Future Enhancements

- Create a new Stock_Batch automatically when a purchase item has no matching batch.
- Send the expiry email only when a batch newly becomes Expired (or another defined alert status).
- Add a scheduled workflow so expiry and status are checked daily without editing a record.
- Verify role-specific access by testing each role, and add record-level criteria where needed.
- Test the remaining status cases and validation cases listed in Table 6.5.
- Deploy through Development, Stage and Production environments with release notes and a backup routine.
- Conduct performance benchmarking with realistic data volumes.

- Integrate with other systems through APIs if required, and extend the platform with pro-code where justified.
- Add AI-based stock forecasting and expiry prediction.
- Develop or enable mobile access.
- Add purchase approval and supplier payment functionality.

9.3 Learning Outcomes and Career Relevance

Developing LabSafe gave practice in data modelling with lookup fields and subforms, writing Deluge validation and update logic, building reports and a dashboard, and documenting and testing an application honestly. These skills relate to roles such as citizen developer, low-code developer and automation engineer.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Status key: **Implemented** = built and (where applicable) tested; **Partial** = partly configured or partly verified; **Unverified** = not confirmed in the project record; **Not implemented**.

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Status	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Implemented	LabSafe application	5.1	Fig. 5.1
II	Forms with ≥ 8 distinct field types	Unverified	Eight forms (confirm field types)	4.3, 5.2	Tables 4.2–4.9
II	Lookup field (relationship between forms)	Implemented	Consumable_Master, Stock_Batch, Stock_Purchase, subforms	4.2, 5.2	Fig. 4.2
II	Subform (parent–child data)	Implemented	Purchase_Items, Issue_Items	4.2, 5.2	Figs. 5.7, 5.9
II	Field validations / layout rules	Partial	Issue validation workflow; field-level rules not verified	5.2, 5.5	Code 5.1
II	List report with filter, grouping, custom action	Partial	Custom and default list reports; filters/actions not verified	5.3	Table 5.2
II	Kanban / Calendar / Timeline / Map report	Not implemented	—	5.3	—
II	Pivot table / chart report	Implemented	Stock_Status_Chart	5.3	Fig. 5.14
II	Page / dashboard built with UI Builder	Implemented	LabSafe_Dashboard	5.4	Fig. 5.15
II	Deluge – form events (validate / success / field actions)	Implemented	Validation and success workflows	5.5	Code 5.1–5.3
II	Deluge – functions, fetch, loops, collections	Partial	Fetch and loops used; standalone functions not documented	5.5	Code 5.2
II	Email / in-app notifications	Implemented	Expiry_Email_Notification (email only)	5.6	Fig. 5.17
II	Form workflow	Implemented	Five form workflows	5.6	Table 5.5
II	Scheduled workflow	Not implemented	—	5.6	—
II	Approval process	Not implemented	—	5.6	—
II	Blueprint	Not implemented	—	5.6	—
II	Batch / report workflow	Not implemented	—	5.6	—
II	Testing & debugging evidence	Implemented	14 tests passed	6	Tables 6.1–6.4
III	Integration with a Zoho service	Not implemented	—	5.7	—
III	Connection / REST API call (invokeurl)	Not implemented	—	5.7	—
III	Webhook or Creator REST API	Not implemented	—	5.7	—

Unit	Zoho Creator Feature	Status	Where in the Application	Report Section	Figure / Code No.
III	Users, roles and permission profiles	Partial	Admin, Lab Staff; enforcement not fully verified	5.8	Fig. 5.18
III	Record-level access criteria / customer portal	Unverified	Not confirmed	5.8	—
III	Zia / AI feature	Not implemented	—	5.9	—
III	Analytics (charts / Zoho Analytics)	Implemented	Chart report and dashboard (Zoho Analytics not used)	5.9	Fig. 5.14
IV	Performance optimisation applied & measured	Not implemented	No benchmarking	7.2	—
IV	Deployment across environments	Unverified	Not verified	7.1	—
IV	Security controls (encryption, OAuth, permissions)	Partial	Roles/permissions only	7.3	Table 7.2
IV	Governance (audit trail, administration)	Unverified	Not documented	7.4	—
V	Pro-code extension	Not implemented	—	5.10	—
V	Low-code limitations & spectrum analysis	Implemented	Documented in report	8.4, 8.5	—
—	Mobile app / portal access	Not implemented	—	5.11	—
—	Backup	Unverified	Not verified	7.4	—

Summary: Implemented – 11; Partial – 5; Unverified – 5; Not implemented – 12; total features listed – 33.

APPENDIX B – COURSE OUTCOME MAPPING

The course outcome statements are those given in the college template. Self-assessed levels are left for the student (1 – Low, 2 – Medium, 3 – High).

Table B.1: Mapping of Course Outcomes to Report Chapters and Project Activities

CO	Course Outcome Statement	Evidenced In (Project Activity)	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2 – justification for low-code, literature, comparison with traditional development	2	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6 – six forms, two subforms, five reports, dashboard, five workflows, Deluge logic, tests	3	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9 – roles configured; no integration implemented (partial evidence only)	2	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7 – deployment approach, security and governance discussed; production deployment not verified	1	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3 – limitations, spectrum reflection	2	

APPENDIX C – COMPLETE DELUGE SOURCE CODE

Important: The exact original Deluge source code was not available when this report was prepared. The listings below are clearly labelled as illustrative examples of the configured logic and must not be read as the implemented source.

C.1 Stock Issue Validation

```
// Script ID : DS-01
// Location  : Stock_Issue -> Validate_Stock_Before_Issue
// Purpose   : Block invalid issue entries
// STATUS    : ILLUSTRATIVE - replace with exact source
for each row in input.Issue_Items
{
    if(row.Batch == null) { alert "Select a batch."; cancel submit;
}
    if(row.Quantity_Issued <= 0) { alert "Quantity must be > 0.";
cancel submit; }
    b = Stock_Batch[ID == row.Batch];
    if(row.Quantity_Issued > b.Available_Quantity) { alert
"Insufficient stock."; cancel submit; }
}
```

C.2 Stock Quantity Deduction

```
// Script ID : DS-02a
// Location  : Stock_Issue -> Update_Stock_After_Issue
// Purpose   : Deduct issued quantity
// STATUS    : ILLUSTRATIVE - replace with exact source
for each row in input.Issue_Items
{
    b = Stock_Batch[ID == row.Batch];
    if(row.Quantity_Issued > 0 && row.Quantity_Issued <=
b.Available_Quantity)
    {
        b.Available_Quantity = b.Available_Quantity -
row.Quantity_Issued;
    }
}
```

C.3 Stock Purchase Quantity Update

```
// Script ID : DS-02b
// Location  : Stock_Purchase -> Update_Stock_After_Purchase
// Purpose   : Increase quantity of an existing matching batch
// STATUS    : ILLUSTRATIVE - replace with exact source
for each item in input.Purchase_Items
{
    b = Stock_Batch[Batch_Number == item.Batch_Number && Consumable
== item.Consumable];
    if(b.count() > 0)
    {
        b.Available_Quantity = b.Available_Quantity +
item.Quantity_Received;
    }
    // new batch is not created automatically
}
```

C.4 Expiry and Stock Status Update

```
// Script ID : DS-03
// Location  : Stock_Batch -> Expiry_Alert
// Purpose   : Set Batch_Status (priority: Depleted, Expired, Low
Stock, Expiring Soon, Available)
// STATUS    : ILLUSTRATIVE - replace with exact source
minLevel = input.Consumable.Minimum_Stock_Level;
today = zoho.currentdate;
if(input.Available_Quantity <= 0) { status = "Depleted"; }
else if(input.Expiry_Date < today) { status = "Expired"; }
else if(input.Available_Quantity <= minLevel) { status = "Low
Stock"; }
else if(input.Expiry_Date <= today.addDay(30)) { status = "Expiring
Soon"; }
else { status = "Available"; }
rec = Stock_Batch[ID == input.ID];
rec.Batch_Status = status;
```

C.5 Email Notification Logic

```
// Script ID : DS-04
// Location  : Stock_Batch -> Expiry_Email_Notification
// Purpose   : Email batch number, expiry date, available quantity,
status
// STATUS    : ILLUSTRATIVE - replace with exact source
sendmail
[
    from : zoho.adminuserid
    to : "[recipient email]"
    subject : "LabSafe batch alert: " + input.Batch_Number
    message : "Batch: " + input.Batch_Number + " | Expiry: " +
input.Expiry_Date + " | Qty: " + input.Available_Quantity + " |
Status: " + input.Batch_Status
]
// configured on any successful add/edit
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

1. Open a web browser (Chrome, Edge or Firefox).
2. Go to the LabSafe URL: https://creatorapp.zoho.in/harikrishnan.ep.2024.cse1207/labsafe#Form:Category_Master
3. Sign in with your Zoho account credentials for your role.
4. The LabSafe home page and menu appear.

D.2 Instructions for Admin

The steps below use the form and report names; where each is located in the menu is confirm from the application.

1. **Add a category:** open Category_Master, enter Category Name, Category Code, Description and Category Status, then submit.
2. **Add a supplier:** open Supplier_Master, enter name, code, contact person, phone, email, address and status, then submit.
3. **Add a consumable:** open Consumable_Master, enter name and code, select the Category and Supplier, enter unit, minimum stock level, storage location, storage conditions and status, then submit.
4. **Create a stock batch:** open Stock_Batch, enter batch number, select the Consumable, enter purchase, manufacturing and expiry dates, initial and available quantity, storage location, then submit. Batch_Status is set automatically.
5. **Record a stock purchase:** open Stock_Purchase, enter date, supplier, invoice number, received by and notes; add rows in Purchase_Items with consumable, batch number, quantity received, dates and unit cost; submit. The quantity of an existing matching batch increases. If no batch matches, create it first in Stock_Batch.
6. **View stock reports:** open Stock_Batch_Report or All_Stock_Batches. Use Stock_Issue_Report for issue history.

7. **View low-stock and expired items:** open Low_Stock_Report and Expired_Items_Report.
8. **Use the dashboard:** open LabSafe_Dashboard to see the stock status chart and stock-related reports.

D.3 Instructions for Lab Staff

Available functions depend on the permissions configured for the role; full enforcement was not verified.

1. **Record a stock issue:** open Stock_Issue, enter issue details (date, issued to, department, purpose, issued by); add rows in Issue_Items selecting consumable, batch, quantity issued and remarks; submit.
2. If a message says stock is insufficient, reduce the quantity or choose another batch.
3. **View reports and the dashboard:** open the stock, low-stock and expired-item reports and LabSafe_Dashboard.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	HARIKRISHNAN EP	**Register No.**	240701164
Project Title	LabSafe – Laboratory Consumable Inventory & Expiry Management System	**Date**	

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded	Remarks
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10		
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10		
3	Forms, Reports & Pages	Variety of field types, lookups/ subforms, report types, UI Builder dashboards	15		
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15		
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10		
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10		
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15		
8	Project Report	Adherence to template and format, clarity, figures, references	5		
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10		
	TOTAL		**100**		

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner Name: _____	**Signature of External Examiner** Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	HARIKRISHNAN EP	**Register No.**	240701164
Project Title	LabSafe – Laboratory Consumable Inventory & Expiry Management System	**Department / Section**	CSE
Application URL Verified	Yes / No	**Date of Viva**	

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiner Comments:

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_____

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INTERNAL EXAMINER	**EXTERNAL EXAMINER**
Name: _____	Name: _____
Signature: _____	Signature: _____